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METHOD AND APPARATUS FOR MEASUREMENT REPORT CONSIDERING HEIGHT IN A WIRELESS COMMUNICATION SYSTEM

Abstract

A method and apparatus for measurement report considering height in a wireless communication system is provided. A wireless device receives information on a cell list associated with a height range. A wireless device monitors a current height of the wireless device. A wireless device sets a measurement report while the current height of the wireless device is in the height range. A wireless device determines whether the measurement report includes measurement results of a first cell based on the cell list; and transmitting the measurement report.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2023/005539, filed on Apr. 24, 2023, which claims the benefit of U.S. Provisional Application No. 63/337,178, filed on May 2, 2022, the contents of which are all incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a method and apparatus for measurement report considering height in a wireless communication system.

BACKGROUND

[0003] 3rd generation partnership project (3GPP) long-term evolution (LTE) is a technology for enabling high-speed packet communications. Many schemes have been proposed for the LTE objective including those that aim to reduce user and provider costs, improve service quality, and expand and improve coverage and system capacity. The 3GPP LTE requires reduced cost per bit, increased service availability, flexible use of a frequency band, a simple structure, an open interface, and adequate power consumption of a terminal as an upper-level requirement.

[0004] Work has started in international telecommunication union (ITU) and 3GPP to develop requirements and specifications for new radio (NR) systems. 3GPP has to identify and develop the technology components needed for successfully standardizing the new RAT timely satisfying both the urgent market needs, and the more long-term requirements set forth by the ITU radio communication sector (ITU-R) international mobile telecommunications (IMT)-2020 process. Further, the NR should be able to use any spectrum band ranging at least up to 100 GHz that may be made available for wireless communications even in a more distant future.

[0005] The NR targets a single technical framework addressing all usage scenarios, requirements and deployment scenarios including enhanced mobile broadband (eMBB), massive machine-type-communications (mMTC), ultra-reliable and low latency communications (URLLC), etc. The NR shall be inherently forward compatible.

SUMMARY

[0006] As the altitude of aerial UEs increases, they experience a line-of-sight propagation condition to more cells, making faraway cells more visible. Consequently, aerial UEs receive interference from more cells in the downlink and cause interference to more cells in the uplink. Additionally, aerial coverage becomes fragmented at higher altitudes. In contrast to terrestrial UEs, which are typically served by the closest network, aerial UEs are served by a side lobe of a neighboring network far from the UE. Therefore, the faraway network may become the serving network, resulting in non-continuous cell coverage.

[0007] As an aerial UE flies across multiple cells at a specific height and direction, the quality of its serving cell fluctuates with a certain frequency, depending on network evaluation. Furthermore, aerial UEs can observe multiple cells with similar signal strength, making it challenging to select a suitable cell based on cell quality alone, such as RSRP and RSRQ. In the case of the measurement report, if an aerial UE detects multiple neighbor cells, it may send measurement reports more frequently, causing more signaling overhead and air interference. To mitigate this issue, the network can configure an excluded list or allowed list of cells to restrict the report. If a cell is on

the excluded cell list, the UE cannot send a measurement report for that cell. If the allowed cell list is configured, the UE can send a measurement report for only cells belonging to the allowed cell list.

[0008] However, to configure the excluded cell list or allowed cell list based on height, the UE should send a measurement report based on the height (H1/H2 event), allowing the network to send a new cell configuration based on the UE's current height. This also results in more signaling overhead between the UE and the network.

[0009] Therefore, studies for measurement report considering height in a wireless communication system are required.

[0010] In an aspect, a method performed by a wireless device in a wireless communication system is provided. The method comprises: receiving information on a cell list associated with a height range; monitoring a current height of the wireless device; setting a measurement report while the current height of the wireless device is in the height range; determining whether the measurement report includes measurement results of a first cell based on the cell list; and transmitting the measurement report.

[0011] In another aspect, an apparatus for implementing the above method is provided.

[0012] The present disclosure can have various advantageous effects.

[0013] According to some embodiments of the present disclosure, a wireless device could efficiently transmit the height-based measurement report by using the height-based cell list.

[0014] For example, if the altitude changes rapidly, the UE can directly perform measurements on the target cell. When the altitude changes in a ping-pong flight mode, it is possible to reduce signaling overhead due to measurement configuration. Additionally, by targeting a network that supports UAVs, measurement reports can be transmitted only about suitable cells at the corresponding height. As a result, unnecessary measurement reports can be reduced, leading to a reduction in UL interference.

[0015] In other words, when the altitude changes rapidly, the UE can directly apply the target cell to perform measurement. When the altitude ping-pongs the flight mode, it is possible to reduce the signaling overhead due to the measurement configuration. Furthermore, by targeting a network that supports UAV, measurement report can be transmitted only about a suitable cell in the corresponding height. Therefore, UL interference can be reduced by reducing unnecessary measurement reports.

[0016] For example, by excluding measurement results for unnecessary cells from the measurement report, the wireless device could save resources.

[0017] According to some embodiments of the present disclosure, a wireless network system could provide an efficient solution for the height-based measurement reporting.

[0018] Advantageous effects which can be obtained through specific embodiments of the present disclosure are not limited to the advantageous effects listed above. For example, there may be a variety of technical effects that a person having ordinary skill in the related art can understand and/or derive from the present disclosure. Accordingly, the specific effects of the present disclosure are not limited to those explicitly described herein, but may include various effects that may be understood or derived from the technical features of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 shows an example of a communication system to which implementations of the present disclosure is applied.

[0020] FIG. 2 shows an example of wireless devices to which implementations of the present disclosure is applied.

[0021] FIG. 3 shows an example of a wireless device to which implementations of the present disclosure is applied.

[0022] FIG. 4 shows another example of wireless devices to which implementations of the present disclosure is applied.

[0023] FIG. 5 shows an example of UE to which implementations of the present disclosure is applied.

[0024] FIGS. 6 and 7 show an example of protocol stacks in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

[0025] FIG. 8 shows a frame structure in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

[0026] FIG. 9 shows a data flow example in the 3GPP NR system to which implementations of the present disclosure is applied.

[0027] FIG. 10 shows an example of a method for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure.

[0028] FIG. 11 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0029] FIG. 12 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0030] FIG. 13 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0031] FIG. 14 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0032] FIG. 15 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0033] FIG. 16 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0034] FIG. 17 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0035] FIG. 18 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0036] FIG. 19 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0037] FIG. 20 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0038] The following techniques, apparatuses, and systems may be applied to a variety of wireless multiple access systems. Examples of the multiple access systems include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, a single carrier frequency division multiple access (SC-FDMA) system, and a multicarrier

frequency division multiple access (MC-FDMA) system. CDMA may be embodied through radio technology such as universal terrestrial radio access (UTRA) or CDMA2000. TDMA may be embodied through radio technology such as global system for mobile communications (GSM), general packet radio service (GPRS), or enhanced data rates for GSM evolution (EDGE). OFDMA may be embodied through radio technology such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, or evolved UTRA (E-UTRA). UTRA is a part of a universal mobile telecommunications system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is a part of evolved UMTS (E-UMTS) using E-UTRA. 3GPP LTE employs OFDMA in DL and SC-FDMA in UL. LTE-advanced (LTE-A) is an evolved version of 3GPP LTE.

[0039] For convenience of description, implementations of the present disclosure are mainly described in regards to a 3GPP based wireless communication system. However, the technical features of the present disclosure are not limited thereto. For example, although the following detailed description is given based on a mobile communication system corresponding to a 3GPP based wireless communication system, aspects of the present disclosure that are not limited to 3GPP based wireless communication system are applicable to other mobile communication systems.

[0040] For terms and technologies which are not specifically described among the terms of and technologies employed in the present disclosure, the wireless communication standard documents published before the present disclosure may be referenced.

[0041] In the present disclosure, “A or B” may mean “only A”, “only B”, or “both A and B”. In other words, “A or B” in the present disclosure may be interpreted as “A and/or B”. For example, “A, B or C” in the present disclosure may mean “only A”, “only B”, “only C”, or “any combination of A, B and C”.

[0042] In the present disclosure, slash (/) or comma (,) may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B or C”.

[0043] In the present disclosure, “at least one of A and B” may mean “only A”, “only B” or “both A and B”. In addition, the expression “at least one of A or B” or “at least one of A and/or B” in the present disclosure may be interpreted as same as “at least one of A and B”.

[0044] In addition, in the present disclosure, “at least one of A, B and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B and C”. In addition, “at least one of A, B or C” or “at least one of A, B and/or C” may mean “at least one of A, B and C”.

[0045] Also, parentheses used in the present disclosure may mean “for example”. In detail, when it is shown as “control information (PDCCH)”, “PDCCH” may be proposed as an example of “control information”. In other words, “control information” in the present disclosure is not limited to “PDCCH”, and “PDCCH” may be proposed as an example of “control information”. In addition, even when shown as “control information (i.e., PDCCH)”, “PDCCH” may be proposed as an example of “control information”.

[0046] Technical features that are separately described in one drawing in the present disclosure may be implemented separately or simultaneously.

[0047] Although not limited thereto, various descriptions, functions, procedures, suggestions, methods and/or operational flowcharts of the present disclosure disclosed herein can be applied to various fields requiring wireless communication and/or connection (e.g., 5G) between devices.

[0048] Hereinafter, the present disclosure will be described in more detail with reference to drawings. The same reference numerals in the following drawings and/or descriptions may refer to the same and/or corresponding hardware blocks, software blocks, and/or functional blocks unless otherwise indicated.

[0049] FIG. 1 shows an example of a communication system to which implementations of the present disclosure is applied.

[0050] The 5G usage scenarios shown in FIG. 1 are only exemplary, and the technical features of the present disclosure can be applied to other 5G usage scenarios which are not shown in FIG. 1.

[0051] Three main requirement categories for 5G include (1) a category of enhanced mobile broadband (eMBB), (2) a category of massive machine type communication (mMTC), and (3) a category of ultra-reliable and low latency communications (URLLC).

[0052] Partial use cases may require a plurality of categories for optimization and other use cases may focus only upon one key performance indicator (KPI). 5G supports such various use cases using a flexible and reliable method.

[0053] eMBB far surpasses basic mobile Internet access and covers abundant bidirectional work and media and entertainment applications in cloud and augmented reality. Data is one of 5G core motive forces and, in a 5G era, a dedicated voice service may not be provided for the first time. In 5G, it is expected that voice will be simply processed as an application program using data connection provided by a communication system. Main causes for increased traffic volume are due to an increase in the size of content and an increase in the number of applications requiring high data transmission rate. A streaming service (of audio and video), conversational video, and mobile Internet access will be more widely used as more devices are connected to the Internet. These many application programs require connectivity of an always turned-on state in order to push real-time information and alarm for users. Cloud storage and applications are rapidly increasing in a mobile communication platform and may be applied to both work and entertainment. The cloud storage is a special use case which accelerates growth of uplink data transmission rate. 5G is also used for remote work of cloud. When a tactile interface is used, 5G demands much lower end-to-end latency to maintain user good experience. Entertainment, for example, cloud gaming and video streaming, is another core element which increases demand for mobile broadband capability. Entertainment is essential for a smartphone and a tablet in any place including high mobility environments such as a train, a vehicle, and an airplane. Other use cases are augmented reality for entertainment and information search. In this case, the augmented reality requires very low latency and instantaneous data volume.

[0054] In addition, one of the most expected 5G use cases relates a function capable of smoothly connecting embedded sensors in all fields, i.e., mMTC. It is expected that the number of potential Internet-of-things (IoT) devices will reach 204 hundred million up to the year of **2020**. An industrial IT is one of categories of performing a main role enabling a smart city, asset tracking, smart utility, agriculture, and security infrastructure through 5G.

[0055] URLLC includes a new service that will change industry through remote control of main infrastructure and an ultra-reliable/available low-latency link such as a self-driving vehicle. A level of reliability and latency is essential to control a smart grid, automatize industry, achieve robotics, and control and adjust a drone.

[0056] 5G is a means of providing streaming evaluated as a few hundred megabits per second to gigabits per second and may complement fiber-to-the-home (FTTH) and cable-based broadband (or DOCSIS). Such fast speed is needed to deliver TV in resolution of 4K or more (6K, 8K, and more), as well as virtual reality and augmented reality. Virtual reality (VR) and augmented reality (AR) applications include almost immersive sports games. A specific application program may require a special network configuration. For example, for VR games, gaming companies need to incorporate a core server into an edge network server of a network operator in order to minimize latency.

[0057] Automotive is expected to be a new important motivated force in 5G together with many use cases for mobile communication for vehicles. For example, entertainment for passengers requires high simultaneous capacity and mobile broadband with high mobility. This is because future users continue to expect connection of high quality regardless of their locations and speeds. Another use case of an automotive field is an AR dashboard. The AR dashboard causes a driver to identify an object in the dark in addition to an object seen from a front window and displays a distance from the object and a movement of the object by overlapping information talking to the

driver. In the future, a wireless module enables communication between vehicles, information exchange between a vehicle and supporting infrastructure, and information exchange between a vehicle and other connected devices (e.g., devices accompanied by a pedestrian). A safety system guides alternative courses of a behavior so that a driver may drive more safely, thereby lowering the danger of an accident. The next stage will be a remotely controlled or self-driven vehicle. This requires very high reliability and very fast communication between different self-driven vehicles and between a vehicle and infrastructure. In the future, a self-driven vehicle will perform all driving activities and a driver will focus only upon abnormal traffic that the vehicle cannot identify. Technical requirements of a self-driven vehicle demand ultra-low latency and ultra-high reliability so that traffic safety is increased to a level that cannot be achieved by human being. [0058] A smart city and a smart home/building mentioned as a smart society will be embedded in a high-density wireless sensor network. A distributed network of an intelligent sensor will identify conditions for costs and energy-efficient maintenance of a city or a home. Similar configurations may be performed for respective households. All of temperature sensors, window and heating controllers, burglar alarms, and home appliances are wirelessly connected. Many of these sensors are typically low in data transmission rate, power, and cost. However, real-time HD video may be demanded by a specific type of device to perform monitoring.

[0059] Consumption and distribution of energy including heat or gas is distributed at a higher level so that automated control of the distribution sensor network is demanded. The smart grid collects information and connects the sensors to each other using digital information and communication technology so as to act according to the collected information. Since this information may include behaviors of a supply company and a consumer, the smart grid may improve distribution of fuels such as electricity by a method having efficiency, reliability, economic feasibility, production sustainability, and automation. The smart grid may also be regarded as another sensor network having low latency.

[0060] Mission critical application (e.g., e-health) is one of 5G use scenarios. A health part contains many application programs capable of enjoying benefit of mobile communication. A communication system may support remote treatment that provides clinical treatment in a faraway place. Remote treatment may aid in reducing a barrier against distance and improve access to medical services that cannot be continuously available in a faraway rural area. Remote treatment is also used to perform important treatment and save lives in an emergency situation. The wireless sensor network based on mobile communication may provide remote monitoring and sensors for parameters such as heart rate and blood pressure.

[0061] Wireless and mobile communication gradually becomes important in the field of an industrial application. Wiring is high in installation and maintenance cost. Therefore, a possibility of replacing a cable with reconstructible wireless links is an attractive opportunity in many industrial fields. However, in order to achieve this replacement, it is necessary for wireless connection to be established with latency, reliability, and capacity similar to those of the cable and management of wireless connection needs to be simplified. Low latency and a very low error probability are new requirements when connection to 5G is needed.

[0062] Logistics and freight tracking are important use cases for mobile communication that enables inventory and package tracking anywhere using a location-based information system. The use cases of logistics and freight typically demand low data rate but require location information with a wide range and reliability.

[0063] Referring to FIG. 1, the communication system 1 includes wireless devices 100a to 100f, base stations (BSs) 200, and a network 300. Although FIG. 1 illustrates a 5G network as an example of the network of the communication system 1, the implementations of the present disclosure are not limited to the 5G system, and can be applied to the future communication system beyond the 5G system.

[0064] The BSs 200 and the network 300 may be implemented as wireless devices and a specific

wireless device may operate as a BS/network node with respect to other wireless devices.

[0065] The wireless devices **100a** to **100f** represent devices performing communication using radio access technology (RAT) (e.g., 5G new RAT (NR)) or LTE) and may be referred to as communication/radio/5G devices. The wireless devices **100a** to **100f** may include, without being limited to, a robot **100a**, vehicles **100b-1** and **100b-2**, an extended reality (XR) device **100c**, a hand-held device **100d**, a home appliance **100e**, an IoT device **100f**, and an artificial intelligence (AI) device/server **400**. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. The vehicles may include an unmanned aerial vehicle (UAV) (e.g., a drone). The XR device may include an AR/VR/Mixed Reality (MR) device and may be implemented in the form of a head-mounted device (HMD), a head-up display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter.

[0066] In the present disclosure, the wireless devices **100a** to **100f** may be called user equipments (UEs). A UE may include, for example, a cellular phone, a smartphone, a laptop computer, a digital broadcast terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation system, a slate personal computer (PC), a tablet PC, an ultrabook, a vehicle, a vehicle having an autonomous traveling function, a connected car, an UAV, an AI module, a robot, an AR device, a VR device, an MR device, a hologram device, a public safety device, an MTC device, an IoT device, a medical device, a FinTech device (or a financial device), a security device, a weather/environment device, a device related to a 5G service, or a device related to a fourth industrial revolution field.

[0067] The UAV may be, for example, an aircraft aviated by a wireless control signal without a human being onboard.

[0068] The VR device may include, for example, a device for implementing an object or a background of the virtual world. The AR device may include, for example, a device implemented by connecting an object or a background of the virtual world to an object or a background of the real world. The MR device may include, for example, a device implemented by merging an object or a background of the virtual world into an object or a background of the real world. The hologram device may include, for example, a device for implementing a stereoscopic image of 360 degrees by recording and reproducing stereoscopic information, using an interference phenomenon of light generated when two laser lights called holography meet.

[0069] The public safety device may include, for example, an image relay device or an image device that is wearable on the body of a user.

[0070] The MTC device and the IoT device may be, for example, devices that do not require direct human intervention or manipulation. For example, the MTC device and the IoT device may include smartmeters, vending machines, thermometers, smartbulbs, door locks, or various sensors.

[0071] The medical device may be, for example, a device used for the purpose of diagnosing, treating, relieving, curing, or preventing disease. For example, the medical device may be a device used for the purpose of diagnosing, treating, relieving, or correcting injury or impairment. For example, the medical device may be a device used for the purpose of inspecting, replacing, or modifying a structure or a function. For example, the medical device may be a device used for the purpose of adjusting pregnancy. For example, the medical device may include a device for treatment, a device for operation, a device for (in vitro) diagnosis, a hearing aid, or a device for procedure.

[0072] The security device may be, for example, a device installed to prevent a danger that may arise and to maintain safety. For example, the security device may be a camera, a closed-circuit TV

(CCTV), a recorder, or a black box.

[0073] The FinTech device may be, for example, a device capable of providing a financial service such as mobile payment. For example, the FinTech device may include a payment device or a point of sales (POS) system.

[0074] The weather/environment device may include, for example, a device for monitoring or predicting a weather/environment.

[0075] The wireless devices **100a** to **100f** may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, a 5G (e.g., NR) network, and a beyond-5G network. Although the wireless devices **100a** to **100f** may communicate with each other through the BSs **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs **200**/network **300**. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g., vehicle-to-vehicle (V2V)/vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

[0076] Wireless communication/connections **150a**, **150b** and **150c** may be established between the wireless devices **100a** to **100f** and/or between wireless device **100a** to **100f** and BS **200** and/or between BSs **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150a**, sidelink communication (or device-to-device (D2D) communication) **150b**, inter-base station communication **150c** (e.g., relay, integrated access and backhaul (IAB)), etc. The wireless devices **100a** to **100f** and the BSs **200**/the wireless devices **100a** to **100f** may transmit/receive radio signals to/from each other through the wireless communication/connections **150a**, **150b** and **150c**. For example, the wireless communication/connections **150a**, **150b** and **150c** may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/de-mapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

[0077] Here, the radio communication technologies implemented in the wireless devices in the present disclosure may include narrowband internet-of-things (NB-IoT) technology for low-power communication as well as LTE, NR and 6G. For example, NB-IoT technology may be an example of low power wide area network (LPWAN) technology, may be implemented in specifications such as LTE Cat NB1 and/or LTE Cat NB2, and may not be limited to the above-mentioned names. Additionally and/or alternatively, the radio communication technologies implemented in the wireless devices in the present disclosure may communicate based on LTE-M technology. For example, LTE-M technology may be an example of LPWAN technology and be called by various names such as enhanced machine type communication (eMTC). For example, LTE-M technology may be implemented in at least one of the various specifications, such as 1) LTE Cat 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-bandwidth limited (non-BL), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and may not be limited to the above-mentioned names. Additionally and/or alternatively, the radio communication technologies implemented in the wireless devices in the present disclosure may include at least one of ZigBee, Bluetooth, and/or LPWAN which take into account low-power communication, and may not be limited to the above-mentioned names. For example, ZigBee technology may generate personal area networks (PANs) associated with small/low-power digital communication based on various specifications such as IEEE 802.15.4 and may be called various names.

[0078] FIG. 2 shows an example of wireless devices to which implementations of the present

disclosure is applied.

[0079] Referring to FIG. 2, a first wireless device **100** and a second wireless device **200** may transmit/receive radio signals to/from an external device through a variety of RATs (e.g., LTE and NR). In FIG. 2, {the first wireless device **100** and the second wireless device **200**} may correspond to at least one of {the wireless device **100a** to **100f** and the BS **200**}, {the wireless device **100a** to **100f** and the wireless device **100a** to ' } and/or {the BS **200** and the BS **200**} of FIG. 1.

[0080] The first wireless device **100** may include one or more processors **102** and one or more memories **104** and additionally further include one or more transceivers **106** and/or one or more antennas **108**. The processor(s) **102** may control the memory(s) **104** and/or the transceiver(s) **106** and may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. For example, the processor(s) **102** may process information within the memory(s) **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) **106**. The processor(s) **102** may receive radio signals including second information/signals through the transceiver(s) **106** and then store information obtained by processing the second information/signals in the memory(s) **104**. The memory(s) **104** may be connected to the processor(s) **102** and may store a variety of information related to operations of the processor(s) **102**. For example, the memory(s) **104** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **102** or for performing the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. Herein, the processor(s) **102** and the memory(s) **104** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **106** may be connected to the processor(s) **102** and transmit and/or receive radio signals through one or more antennas **108**. Each of the transceiver(s) **106** may include a transmitter and/or a receiver. The transceiver(s) **106** may be interchangeably used with radio frequency (RF) unit(s). In the present disclosure, the first wireless device **100** may represent a communication modem/circuit/chip.

[0081] The second wireless device **200** may include one or more processors **202** and one or more memories **204** and additionally further include one or more transceivers **206** and/or one or more antennas **208**. The processor(s) **202** may control the memory(s) **204** and/or the transceiver(s) **206** and may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. For example, the processor(s) **202** may process information within the memory(s) **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) **206**. The processor(s) **202** may receive radio signals including fourth information/signals through the transceiver(s) **106** and then store information obtained by processing the fourth information/signals in the memory(s) **204**. The memory(s) **204** may be connected to the processor(s) **202** and may store a variety of information related to operations of the processor(s) **202**. For example, the memory(s) **204** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **202** or for performing the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts described in the present disclosure. Herein, the processor(s) **202** and the memory(s) **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **206** may be connected to the processor(s) **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver(s) **206** may include a transmitter and/or a receiver. The transceiver(s) **206** may be interchangeably used with RF unit(s). In the present disclosure, the second wireless device **200** may represent a communication modem/circuit/chip.

[0082] Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may

implement one or more layers (e.g., functional layers such as physical (PHY) layer, media access control (MAC) layer, radio link control (RLC) layer, packet data convergence protocol (PDCP) layer, radio resource control (RRC) layer, and service data adaptation protocol (SDAP) layer). The one or more processors **102** and **202** may generate one or more protocol data units (PDUs) and/or one or more service data unit (SDUs) according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure.

[0083] The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more application specific integrated circuits (ASICs), one or more digital signal processors (DSPs), one or more digital signal processing devices (DSPDs), one or more programmable logic devices (PLDs), or one or more field programmable gate arrays (FPGAs) may be included in the one or more processors **102** and **202**. descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

[0084] The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by read-only memories (ROMs), random access memories (RAMs), electrically erasable programmable read-only memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

[0085] The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more

processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices.

[0086] The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure, through the one or more antennas **108** and **208**. In the present disclosure, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports).

[0087] The one or more transceivers **106** and **206** may convert received radio signals/channels, etc., from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc., using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc., processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters. For example, the transceivers **106** and **206** can up-convert OFDM baseband signals to a carrier frequency by their (analog) oscillators and/or filters under the control of the processors **102** and **202** and transmit the up-converted OFDM signals at the carrier frequency. The transceivers **106** and **206** may receive OFDM signals at a carrier frequency and down-convert the OFDM signals into OFDM baseband signals by their (analog) oscillators and/or filters under the control of the transceivers **102** and **202**.

[0088] In the implementations of the present disclosure, a UE may operate as a transmitting device in uplink (UL) and as a receiving device in downlink (DL). In the implementations of the present disclosure, a BS may operate as a receiving device in UL and as a transmitting device in DL. Hereinafter, for convenience of description, it is mainly assumed that the first wireless device **100** acts as the UE, and the second wireless device **200** acts as the BS. For example, the processor(s) **102** connected to, mounted on or launched in the first wireless device **100** may be configured to perform the UE behavior according to an implementation of the present disclosure or control the transceiver(s) **106** to perform the UE behavior according to an implementation of the present disclosure. The processor(s) **202** connected to, mounted on or launched in the second wireless device **200** may be configured to perform the BS behavior according to an implementation of the present disclosure or control the transceiver(s) **206** to perform the BS behavior according to an implementation of the present disclosure.

[0089] In the present disclosure, a BS is also referred to as a node B (NB), an eNode B (eNB), or a gNB.

[0090] FIG. 3 shows an example of a wireless device to which implementations of the present disclosure is applied.

[0091] The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 1).

[0092] Referring to FIG. 3, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 2 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit **110** may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** of FIG. 2 and/or the one or more memories **104** and **204** of FIG. 2. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** of FIG. 2 and/or the one or more antennas **108** and **208** of FIG. 2. The control unit **120** is electrically connected to the

communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of each of the wireless devices **100** and **200**. For example, the control unit **120** may control an electric/mechanical operation of each of the wireless devices **100** and **200** based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

[0093] The additional components **140** may be variously configured according to types of the wireless devices **100** and **200**. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit (e.g., audio I/O port, video I/O port), a driving unit, and a computing unit. The wireless devices **100** and **200** may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 1), the vehicles (**100b-1** and **100b-2** of FIG. 1), the XR device (**100c** of FIG. 1), the hand-held device (**100d** of FIG. 1), the home appliance (**100e** of FIG. 1), the IoT device (**100f** of FIG. 1), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a FinTech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. 1), the BSs (**200** of FIG. 1), a network node, etc. The wireless devices **100** and **200** may be used in a mobile or fixed place according to a use-example/service.

[0094] In FIG. 3, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor (AP), an electronic control unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a RAM, a DRAM, a ROM, a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

[0095] FIG. 4 shows another example of wireless devices to which implementations of the present disclosure is applied.

[0096] Referring to FIG. 4, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 2 and may be configured by various elements, components, units/portions, and/or modules.

[0097] The first wireless device **100** may include at least one transceiver, such as a transceiver **106**, and at least one processing chip, such as a processing chip **101**. The processing chip **101** may include at least one processor, such a processor **102**, and at least one memory, such as a memory **104**. The memory **104** may be operably connectable to the processor **102**. The memory **104** may store various types of information and/or instructions. The memory **104** may store a software code **105** which implements instructions that, when executed by the processor **102**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **105** may implement instructions that, when executed by the processor **102**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **105** may control the processor **102** to perform one or more protocols. For example, the software code **105** may control the processor **102** may perform one or more layers of the radio interface protocol.

[0098] The second wireless device **200** may include at least one transceiver, such as a transceiver **206**, and at least one processing chip, such as a processing chip **201**. The processing chip **201** may include at least one processor, such as a processor **202**, and at least one memory, such as a memory **204**. The memory **204** may be operably connectable to the processor **202**. The memory **204** may store various types of information and/or instructions. The memory **204** may store a software code **205** which implements instructions that, when executed by the processor **202**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **205** may implement instructions that, when executed by the processor **202**, perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. For example, the software code **205** may control the processor **202** to perform one or more protocols. For example, the software code **205** may control the processor **202** may perform one or more layers of the radio interface protocol.

[0099] FIG. 5 shows an example of UE to which implementations of the present disclosure is applied.

[0100] Referring to FIG. 5, a UE **100** may correspond to the first wireless device **100** of FIG. 2 and/or the first wireless device **100** of FIG. 4.

[0101] A UE **100** includes a processor **102**, a memory **104**, a transceiver **106**, one or more antennas **108**, a power management module **110**, a battery **112**, a display **114**, a keypad **116**, a subscriber identification module (SIM) card **118**, a speaker **120**, and a microphone **122**.

[0102] The processor **102** may be configured to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The processor **102** may be configured to control one or more other components of the UE **100** to implement the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. Layers of the radio interface protocol may be implemented in the processor **102**. The processor **102** may include ASIC, other chipset, logic circuit and/or data processing device. The processor **102** may be an application processor. The processor **102** may include at least one of a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), a modem (modulator and demodulator). An example of the processor **102** may be found in SNAPDRAGON™ series of processors made by Qualcomm®, EXYNOS™ series of processors made by Samsung®, A series of processors made by Apple®, HELIO™ series of processors made by MediaTek®, ATOM™ series of processors made by Intel® or a corresponding next generation processor.

[0103] The memory **104** is operatively coupled with the processor **102** and stores a variety of information to operate the processor **102**. The memory **104** may include ROM, RAM, flash memory, memory card, storage medium and/or other storage device. When the embodiments are implemented in software, the techniques described herein can be implemented with modules (e.g., procedures, functions, etc.) that perform the descriptions, functions, procedures, suggestions, methods and/or operational flowcharts disclosed in the present disclosure. The modules can be stored in the memory **104** and executed by the processor **102**. The memory **104** can be implemented within the processor **102** or external to the processor **102** in which case those can be communicatively coupled to the processor **102** via various means as is known in the art.

[0104] The transceiver **106** is operatively coupled with the processor **102**, and transmits and/or receives a radio signal. The transceiver **106** includes a transmitter and a receiver. The transceiver **106** may include baseband circuitry to process radio frequency signals. The transceiver **106** controls the one or more antennas **108** to transmit and/or receive a radio signal.

[0105] The power management module **110** manages power for the processor **102** and/or the transceiver **106**. The battery **112** supplies power to the power management module **110**.

[0106] The display **114** outputs results processed by the processor **102**. The keypad **116** receives inputs to be used by the processor **102**. The keypad **16** may be shown on the display **114**.

[0107] The SIM card **118** is an integrated circuit that is intended to securely store the international mobile subscriber identity (IMSI) number and its related key, which are used to identify and authenticate subscribers on mobile telephony devices (such as mobile phones and computers). It is also possible to store contact information on many SIM cards.

[0108] The speaker **120** outputs sound-related results processed by the processor **102**. The microphone **122** receives sound-related inputs to be used by the processor **102**.

[0109] FIGS. **6** and **7** show an example of protocol stacks in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

[0110] In particular, FIG. **6** illustrates an example of a radio interface user plane protocol stack between a UE and a BS and FIG. **7** illustrates an example of a radio interface control plane protocol stack between a UE and a BS. The control plane refers to a path through which control messages used to manage call by a UE and a network are transported. The user plane refers to a path through which data generated in an application layer, for example, voice data or Internet packet data are transported. Referring to FIG. **6**, the user plane protocol stack may be divided into Layer 1 (i.e., a PHY layer) and Layer 2. Referring to FIG. **7**, the control plane protocol stack may be divided into Layer 1 (i.e., a PHY layer), Layer 2, Layer 3 (e.g., an RRC layer), and a non-access stratum (NAS) layer. Layer 1, Layer 2 and Layer 3 are referred to as an access stratum (AS).

[0111] In the 3GPP LTE system, the Layer 2 is split into the following sublayers: MAC, RLC, and PDCP. In the 3GPP NR system, the Layer 2 is split into the following sublayers: MAC, RLC, PDCP and SDAP. The PHY layer offers to the MAC sublayer transport channels, the MAC sublayer offers to the RLC sublayer logical channels, the RLC sublayer offers to the PDCP sublayer RLC channels, the PDCP sublayer offers to the SDAP sublayer radio bearers. The SDAP sublayer offers to 5G core network quality of service (QoS) flows.

[0112] In the 3GPP NR system, the main services and functions of the MAC sublayer include: mapping between logical channels and transport channels; multiplexing/de-multiplexing of MAC SDUs belonging to one or different logical channels into/from transport blocks (TB) delivered to/from the physical layer on transport channels; scheduling information reporting; error correction through hybrid automatic repeat request (HARQ) (one HARQ entity per cell in case of carrier aggregation (CA)); priority handling between UEs by means of dynamic scheduling; priority handling between logical channels of one UE by means of logical channel prioritization; padding. A single MAC entity may support multiple numerologies, transmission timings and cells. Mapping restrictions in logical channel prioritization control which numerology (ies), cell(s), and transmission timing(s) a logical channel can use.

[0113] Different kinds of data transfer services are offered by MAC. To accommodate different kinds of data transfer services, multiple types of logical channels are defined, i.e., each supporting transfer of a particular type of information. Each logical channel type is defined by what type of information is transferred. Logical channels are classified into two groups: control channels and traffic channels. Control channels are used for the transfer of control plane information only, and traffic channels are used for the transfer of user plane information only. Broadcast control channel (BCCH) is a downlink logical channel for broadcasting system control information, paging control channel (PCCH) is a downlink logical channel that transfers paging information, system information change notifications and indications of ongoing public warning service (PWS) broadcasts, common control channel (CCCH) is a logical channel for transmitting control information between UEs and network and used for UEs having no RRC connection with the network, and dedicated control channel (DCCH) is a point-to-point bi-directional logical channel that transmits dedicated control information between a UE and the network and used by UEs having an RRC connection. Dedicated traffic channel (DTCH) is a point-to-point logical channel, dedicated to one UE, for the transfer of user information. A DTCH can exist in both uplink and downlink. In downlink, the following connections between logical channels and transport channels exist: BCCH can be mapped to broadcast channel (BCH); BCCH can be mapped to downlink

shared channel (DL-SCH); PCCH can be mapped to paging channel (PCH); CCCH can be mapped to DL-SCH; DCCH can be mapped to DL-SCH; and DTCH can be mapped to DL-SCH. In uplink, the following connections between logical channels and transport channels exist: CCCH can be mapped to uplink shared channel (UL-SCH); DCCH can be mapped to UL-SCH; and DTCH can be mapped to UL-SCH.

[0114] The RLC sublayer supports three transmission modes: transparent mode (TM), unacknowledged mode (UM), and acknowledged mode (AM). The RLC configuration is per logical channel with no dependency on numerologies and/or transmission durations. In the 3GPP NR system, the main services and functions of the RLC sublayer depend on the transmission mode and include: transfer of upper layer PDUs; sequence numbering independent of the one in PDCP (UM and AM); error correction through ARQ (AM only); segmentation (AM and UM) and re-segmentation (AM only) of RLC SDUs; reassembly of SDU (AM and UM); duplicate detection (AM only); RLC SDU discard (AM and UM); RLC re-establishment; protocol error detection (AM only).

[0115] In the 3GPP NR system, the main services and functions of the PDCP sublayer for the user plane include: sequence numbering; header compression and decompression using robust header compression (ROHC); transfer of user data; reordering and duplicate detection; in-order delivery; PDCP PDU routing (in case of split bearers); retransmission of PDCP SDUs; ciphering, deciphering and integrity protection; PDCP SDU discard; PDCP re-establishment and data recovery for RLC AM; PDCP status reporting for RLC AM; duplication of PDCP PDUs and duplicate discard indication to lower layers. The main services and functions of the PDCP sublayer for the control plane include: sequence numbering; ciphering, deciphering and integrity protection; transfer of control plane data; reordering and duplicate detection; in-order delivery; duplication of PDCP PDUs and duplicate discard indication to lower layers.

[0116] In the 3GPP NR system, the main services and functions of SDAP include: mapping between a QoS flow and a data radio bearer; marking QoS flow ID (QFI) in both DL and UL packets. A single protocol entity of SDAP is configured for each individual PDU session.

[0117] In the 3GPP NR system, the main services and functions of the RRC sublayer include: broadcast of system information related to AS and NAS; paging initiated by 5GC or NG-RAN; establishment, maintenance and release of an RRC connection between the UE and NG-RAN; security functions including key management; establishment, configuration, maintenance and release of signaling radio bearers (SRBs) and data radio bearers (DRBs); mobility functions (including: handover and context transfer, UE cell selection and reselection and control of cell selection and reselection, inter-RAT mobility); QoS management functions; UE measurement reporting and control of the reporting; detection of and recovery from radio link failure; NAS message transfer to/from NAS from/to UE.

[0118] FIG. 8 shows a frame structure in a 3GPP based wireless communication system to which implementations of the present disclosure is applied.

[0119] The frame structure shown in FIG. 8 is purely exemplary and the number of subframes, the number of slots, and/or the number of symbols in a frame may be variously changed. In the 3GPP based wireless communication system, OFDM numerologies (e.g., subcarrier spacing (SCS), transmission time interval (TTI) duration) may be differently configured between a plurality of cells aggregated for one UE. For example, if a UE is configured with different SCSs for cells aggregated for the cell, an (absolute time) duration of a time resource (e.g., a subframe, a slot, or a TTI) including the same number of symbols may be different among the aggregated cells. Herein, symbols may include OFDM symbols (or CP-OFDM symbols), SC-FDMA symbols (or discrete Fourier transform-spread-OFDM (DFT-s-OFDM) symbols).

[0120] Referring to FIG. 8, downlink and uplink transmissions are organized into frames. Each frame has $T_{sub.f}=10$ ms duration. Each frame is divided into two half-frames, where each of the half-frames has 5 ms duration. Each half-frame consists of 5 subframes, where the duration $T_{sub.sf}$

per subframe is 1 ms. Each subframe is divided into slots and the number of slots in a subframe depends on a subcarrier spacing. Each slot includes 14 or 12 OFDM symbols based on a cyclic prefix (CP). In a normal CP, each slot includes 14 OFDM symbols and, in an extended CP, each slot includes 12 OFDM symbols. The numerology is based on exponentially scalable subcarrier spacing $\Delta f = 2^{\sup.u} \cdot 15$ kHz.

[0121] Table 1 shows the number of OFDM symbols per slot $N_{\sup.slot.sub.symb}$, the number of slots per frame $N_{\sup.frame,u.sub.slot}$, and the number of slots per subframe $N_{\sup.subframe,u.sub.slot}$ for the normal CP, according to the subcarrier spacing $\Delta f = 2^{\sup.u} \cdot 15$ kHz.

TABLE-US-00001 TABLE 1 u $N_{\sup.slot.sub.symb}$ $N_{\sup.frame}$ u $N_{\sup.subframe}$ u $N_{\sup.subframe,u.sub.slot}$

0	14	10	1	1	14	20	2	2	14	40	4	3	14	80	8	4	14	160	16
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[0122] Table 2 shows the number of OFDM symbols per slot $N_{\sup.slot.sub.symb}$, the number of slots per frame $N_{\sup.frame,u.sub.slot}$, and the number of slots per subframe $N_{\sup.subframe,u.sub.slot}$ for the extended CP, according to the subcarrier spacing $\Delta f = 2^{\sup.u} \cdot 15$ kHz.

TABLE-US-00002 TABLE 2 u $N_{\sup.slot.sub.symb}$ $N_{\sup.frame}$ u $N_{\sup.subframe}$ u $N_{\sup.subframe,u.sub.slot}$

2	12	40	4
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[0123] A slot includes plural symbols (e.g., 14 or 12 symbols) in the time domain. For each numerology (e.g., subcarrier spacing) and carrier, a resource grid of $N_{\sup.size,u.sub.grid,x} \cdot N_{\sup.RB.sub.sc}$ subcarriers and $N_{\sup.subframe,u.sub.symb}$ OFDM symbols is defined, starting at common resource block (CRB) $N_{\sup.start,u.sub.grid}$ indicated by higher-layer signaling (e.g., RRC signaling), where $N_{\sup.size,u.sub.grid,x}$ is the number of resource blocks (RBs) in the resource grid and the subscript x is DL for downlink and UL for uplink. $N_{\sup.RB.sub.sc}$ is the number of subcarriers per RB. In the 3GPP based wireless communication system, $N_{\sup.RB.sub.sc}$ is 12 generally. There is one resource grid for a given antenna port p , subcarrier spacing configuration u , and transmission direction (DL or UL). The carrier bandwidth $N_{\sup.size,u.sub.grid}$ for subcarrier spacing configuration u is given by the higher-layer parameter (e.g., RRC parameter). Each element in the resource grid for the antenna port p and the subcarrier spacing configuration u is referred to as a resource element (RE) and one complex symbol may be mapped to each RE. Each RE in the resource grid is uniquely identified by an index k in the frequency domain and an index/representing a symbol location relative to a reference point in the time domain. In the 3GPP based wireless communication system, an RB is defined by 12 consecutive subcarriers in the frequency domain.

[0124] In the 3GPP NR system, RBs are classified into CRBs and physical resource blocks (PRBs). CRBs are numbered from 0 and upwards in the frequency domain for subcarrier spacing configuration u . The center of subcarrier 0 of CRB 0 for subcarrier spacing configuration u coincides with ‘point A’ which serves as a common reference point for resource block grids. In the 3GPP NR system, PRBs are defined within a bandwidth part (BWP) and numbered from 0 to $N_{\sup.size.sub.BWP,i}-1$, where i is the number of the bandwidth part. The relation between the physical resource block $n_{sub.PRb}$ in the bandwidth part i and the common resource block $n_{sub.CRB}$ is as follows: $n_{sub.PRb} = n_{sub.CRB} + N_{\sup.size.sub.BWP,i}$, where $N_{\sup.size.sub.BWP,i}$ is the common resource block where bandwidth part starts relative to CRB 0. The BWP includes a plurality of consecutive RBs. A carrier may include a maximum of N (e.g., 5) BWPs. A UE may be configured with one or more BWPs on a given component carrier. Only one BWP among BWPs configured to the UE can active at a time. The active BWP defines the UE's operating bandwidth within the cell's operating bandwidth.

[0125] The NR frequency band may be defined as two types of frequency range, i.e., FR1 and FR2. The numerical value of the frequency range may be changed. For example, the frequency ranges of the two types (FR1 and FR2) may be as shown in Table 3 below. For ease of explanation, in the frequency ranges used in the NR system, FR1 may mean “sub 6 GHz range”, FR2 may mean

“above 6 GHz range,” and may be referred to as millimeter wave (mmW).

TABLE-US-00003 TABLE 3 Frequency Range Corresponding frequency designation range
Subcarrier Spacing FR1 450 MHz-6000 MHz 15, 30, 60 kHz FR2 24250 MHz-52600 MHz 60, 120, 240 kHz

[0126] As mentioned above, the numerical value of the frequency range of the NR system may be changed. For example, FR1 may include a frequency band of 410 MHz to 7125 MHz as shown in Table 4 below. That is, FR1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, etc.) or more. For example, a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, etc.) or more included in FR1 may include an unlicensed band. Unlicensed bands may be used for a variety of purposes, for example for communication for vehicles (e.g., autonomous driving).

TABLE-US-00004 TABLE 4 Frequency Range Corresponding frequency designation range
Subcarrier Spacing FR1 410 MHz-7125 MHz 15, 30, 60 kHz FR2 24250 MHz-52600 MHz 60, 120, 240 kHz

[0127] In the present disclosure, the term “cell” may refer to a geographic area to which one or more nodes provide a communication system, or refer to radio resources. A “cell” as a geographic area may be understood as coverage within which a node can provide service using a carrier and a “cell” as radio resources (e.g., time-frequency resources) is associated with bandwidth which is a frequency range configured by the carrier. The “cell” associated with the radio resources is defined by a combination of downlink resources and uplink resources, for example, a combination of a DL component carrier (CC) and a UL CC. The cell may be configured by downlink resources only, or may be configured by downlink resources and uplink resources. Since DL coverage, which is a range within which the node is capable of transmitting a valid signal, and UL coverage, which is a range within which the node is capable of receiving the valid signal from the UE, depends upon a carrier carrying the signal, the coverage of the node may be associated with coverage of the “cell” of radio resources used by the node. Accordingly, the term “cell” may be used to represent service coverage of the node sometimes, radio resources at other times, or a range that signals using the radio resources can reach with valid strength at other times.

[0128] In CA, two or more CCs are aggregated. A UE may simultaneously receive or transmit on one or multiple CCs depending on its capabilities. CA is supported for both contiguous and non-contiguous CCs. When CA is configured, the UE only has one RRC connection with the network. At RRC connection establishment/re-establishment/handover, one serving cell provides the NAS mobility information, and at RRC connection re-establishment/handover, one serving cell provides the security input. This cell is referred to as the primary cell (PCell). The PCell is a cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure. Depending on UE capabilities, secondary cells (SCells) can be configured to form together with the PCell a set of serving cells. An SCell is a cell providing additional radio resources on top of special cell (SpCell). The configured set of serving cells for a UE therefore always consists of one PCell and one or more SCells. For dual connectivity (DC) operation, the term SpCell refers to the PCell of the master cell group (MCG) or the primary SCell (PSCell) of the secondary cell group (SCG). An SpCell supports PUCCH transmission and contention-based random access, and is always activated. The MCG is a group of serving cells associated with a master node, comprised of the SpCell (PCell) and optionally one or more SCells. The SCG is the subset of serving cells associated with a secondary node, comprised of the PSCell and zero or more SCells, for a UE configured with DC. For a UE in RRC_CONNECTED not configured with CA/DC, there is only one serving cell comprised of the PCell. For a UE in RRC_CONNECTED configured with CA/DC, the term “serving cells” is used to denote the set of cells comprised of the SpCell(s) and all SCells. In DC, two MAC entities are configured in a UE: one for the MCG and one for the SCG.

[0129] FIG. 9 shows a data flow example in the 3GPP NR system to which implementations of the present disclosure is applied.

[0130] Referring to FIG. 9, “RB” denotes a radio bearer, and “H” denotes a header. Radio bearers are categorized into two groups: DRBs for user plane data and SRBs for control plane data. The MAC PDU is transmitted/received using radio resources through the PHY layer to/from an external device. The MAC PDU arrives to the PHY layer in the form of a transport block.

[0131] In the PHY layer, the uplink transport channels UL-SCH and RACH are mapped to their physical channels PUSCH and PRACH, respectively, and the downlink transport channels DL-SCH, BCH and PCH are mapped to PDSCH, PBCH and PDSCH, respectively. In the PHY layer, uplink control information (UCI) is mapped to PUCCH, and downlink control information (DCI) is mapped to PDCCH. A MAC PDU related to UL-SCH is transmitted by a UE via a PUSCH based on an UL grant, and a MAC PDU related to DL-SCH is transmitted by a BS via a PDSCH based on a DL assignment.

[0132] Hereinafter, technical features related to the flight path are described. Parts of section 5.3.3.4, 5.5.4.16, 5.5.4.17, 5.6.5.3, and 6.2.2 of 3GPP TS 36.331 v16.6.0 may be referred.

[0133] Operations related to reception of the RRCConnectionSetup by the UE are described. The UE shall: [0134] 1> set the content of RRCConnectionSetupComplete message as follows: [0135] 2> if the UE is connected to EPC: [0136] 3> except for NB-IoT: [0137] 4> include the mobilityState and set it to the mobility state of the UE just prior to entering RRC_CONNECTED state; [0138] 4> if the UE has flight path information available: [0139] 5> include flightPathInfoAvailable;

[0140] Operations related to reception of the UEInformationRequest message are described. [0141] 1> except for NB-IoT, if flightPathInfoReq field is present and the UE has flight path information available: [0142] 2> include the flightPathInfoReport and set it to include the list of waypoints along the flight path; [0143] 2> if the includeTimeStamp is set to TRUE: [0144] 3> set the field timeStamp to the time when UE intends to arrive to each waypoint if this information is available at the UE;

[0145] Technical features related to a UEInformationRequest message are described. The UEInformationRequest is the command used by E-UTRAN to retrieve information from the UE.

[0146] For example, signalling radio bearer for the UEInformationRequest may include SRB1. RLC-Service Access Point (SAP) for the UEInformationRequest may include AM. Logical channel for the UEInformationRequest may include DCCH. Direction for the UEInformationRequest may be E-UTRAN to UE.

[0147] The UEInformationRequest may include information on a flightPathInfoReq (for example, FlightPathInfoReportConfig) and/or information on nonCriticalExtension.

[0148] Technical features related to a UEInformationResponse message are described. For example, the UEInformationResponse message is used by the UE to transfer the information requested by the E-UTRAN.

[0149] For example, signalling radio bearer for the UEInformationResponse may include SRB1 or SRB2 (when logged measurement information is included). RLC-SAP for the UEInformationResponse may include an AM. Logical channel for the UEInformationResponse may include a DCCH. Direction for the UEInformationResponse may be UE to E-UTRAN.

[0150] For example, UEInformationResponse message may include a flightPathInfoReport. For example, the flightPathInfoReport may include information on one or more flightPaths and/or one or more wayPointLocations.

[0151] Technical features related to LocationInfo are described. For example, the IE LocationInfo is used to transfer detailed location information available at the UE to correlate measurements and UE position information.

[0152] For example, LocationInfo information element may include verticalVelocityInfo including information on a verticalVelocity and a verticalVelocity AndUncertainty.

[0153] For example, a verticalVelocity AndUncertainty may include information on a parameter vertical Velocity AndUncertainty corresponds to horizontalWithVerticalVelocity AndUncertainty.

The first/leftmost bit of the first octet contains the most significant bit.

[0154] For example, a verticalVelocity may include information on a parameter verticalVelocity corresponds to horizontal With VerticalVelocity. The first/leftmost bit of the first octet contains the most significant bit.

[0155] UE operations related to Event H1 (The Aerial UE height is above a threshold) are described.

[0156] The UE shall: [0157] 1> consider the entering condition for this event to be satisfied when condition H1-1, as specified below, is fulfilled; [0158] 1> consider the leaving condition for this event to be satisfied when condition H1-2, as specified below, is fulfilled;

$Ms - Hys > Thresh + Offset$ Inequality H1-1 (Entering Condition)

$Ms + Hys < Thresh + Offset$ Inequality H1-2 (Leaving Condition)

[0159] The variables in the formula are defined as follows: [0160] Ms is the Aerial UE height, not taking into account any offsets. [0161] Hys is the hysteresis parameter (i.e. h1-Hysteresis as defined within ReportConfigEUTRA) for this event. [0162] Thresh is the reference threshold parameter for this event given in MeasConfig (i.e. heightThreshRef as defined within MeasConfig). [0163] Offset is the offset value to heightThreshRef to obtain the absolute threshold for this event. (i.e. h1-ThresholdOffset as defined within ReportConfigEUTRA) [0164] Ms is expressed in meters. [0165] Thresh is expressed in the same unit as Ms.

[0166] UE operations related to Event H2 (The Aerial UE height is below a threshold) are described.

[0167] The UE shall: [0168] 1> consider the entering condition for this event to be satisfied when condition H2-1, as specified below, is fulfilled; [0169] 1> consider the leaving condition for this event to be satisfied when condition H2-2, as specified below, is fulfilled;

[00001] Inequality H2 - 1 (Entering condition) $Ms + Hys < Thresh + Offset$

Inequality H2 - 2 (Leaving condition) $Ms + Hys > Thresh + Offset$

[0170] The variables in the formula are defined as follows: [0171] Ms is the Aerial UE height, not taking into account any offsets. [0172] Hys is the hysteresis parameter (i.e. h2-Hysteresis as defined within ReportConfigEUTRA) for this event. [0173] Thresh is the reference threshold parameter for this event given in MeasConfig (i.e. heightThreshRef as defined within MeasConfig). [0174] Offset is the offset value to heightThreshRef to obtain the absolute threshold for this event. (i.e. h2-ThresholdOffset as defined within ReportConfigEUTRA) [0175] Ms is expressed in meters. [0176] Thresh is expressed in the same unit as Ms.

[0177] Hereinafter, technical features related to Aerial UE communication are described. Parts of section 23.17 of 3GPP TS 36.300 v16.5.0 may be referred.

[0178] E-UTRAN based mechanisms providing LTE connection to UEs capable of Aerial communication are supported via the following functionalities: [0179] subscription-based Aerial UE identification and authorization. [0180] height reporting based on the event that the UE's altitude has crossed a network-configured reference altitude threshold. [0181] interference detection based on a measurement reporting that is triggered when a configured number of cells (i.e. larger than one) fulfills the triggering criteria simultaneously. [0182] signalling of flight path information from UE to E-UTRAN. [0183] Location information reporting, including UE's horizontal and vertical velocity.

[Subscription Based Identification of Aerial UE Function]

[0184] Support of Aerial UE function is stored in the user's subscription information in HSS. HSS transfers this information to the MME during Attach, Service Request and Tracking Area Update procedures.

[0185] The subscription information can be provided from the MME to the eNB via the S1 AP Initial Context Setup Request during Attach, Tracking Area Update and Service Request

procedures. In addition, for X2-based handover, the source eNodeB can include the subscription information in the X2-AP Handover Request message to the target eNodeB.

[0186] For the intra and inter MME S1 based handover, the MME provides the subscription information to the target eNB after the handover procedure.

[Height Based Reporting for Aerial UE Communication]

[0187] An aerial UE can be configured with event based height reporting. UE sends height report when the altitude of the aerial UE is above or below a configured threshold. The report contains height and location if configured.

[Interference Detection and Mitigation for Aerial UE Communication]

[0188] For interference detection, an aerial UE can be configured with RRM event A3, A4 or A5 that triggers measurement report when individual (per cell) RSRP values for a configured number of cells fulfil the configured event. The report contains RRM results and location if configured.

[0189] For interference mitigation an aerial UE can be configured with a dedicated UE-specific alpha parameter for PUSCH power control.

[Flight Path Information Reporting]

[0190] E-UTRAN can request a UE to report flight path information consisting of a number of waypoints defined as 3D locations. A UE reports up to configured number of waypoints if flight path information is available at the UE. The report can consist also time stamps per waypoint if configured in the request and if available at the UE.

[Location Reporting for Aerial UE Communication]

[0191] Location information for Aerial UE communication can include horizontal and vertical speed if configured. Location information can be included in RRM report and in height report.

[0192] Hereinafter, technical features related to measurement report triggering are described. Parts of section 5.5.4 of 3GPP TS 36.331 v16.6.0 may be referred. [0193] 2> if the triggerType is set to event, and if the corresponding reportConfig does not include numberOfTriggeringCells, and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable cells for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig, while the VarMeasReportList does not include a measurement reporting entry for this measId (a first cell triggers the event): [0194] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0195] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0196] 3> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0197] 3> if the UE supports T312 and if useT312 is set to true for this event and if T310 is running: [0198] 4> if T312 is not running: [0199] 5> start timer T312 with the value configured in the corresponding measObject; [0200] 3> initiate the measurement reporting procedure; [0201] 2> if the triggerType is set to event, and if the corresponding reportConfig does not include numberOfTriggeringCells, and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable cells not included in the cellsTriggeredList for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig (a subsequent cell triggers the event): [0202] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0203] 3> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0204] 3> if the UE supports T312 and if useT312 is set to true for this event and if T310 is running: [0205] 4> if T312 is not running: [0206] 5> start timer T312 with the value configured in the corresponding measObject; [0207] 3> initiate the measurement reporting procedure; [0208] 2> if the triggerType is set to event and if the corresponding reportConfig includes numberOfTriggeringCells, and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable cells for all measurements after layer 3

filtering taken during timeToTrigger defined for this event within the VarMeasConfig: [0209] 3> If the VarMeasReportList does not include a measurement reporting entry for this measId (a first cell triggers the event): [0210] 4> include a measurement reporting entry within the VarMeasReportList for this measId; [0211] 3> If the number of cell(s) in the cellsTriggeredList is larger than or equal to numberOfTriggeringCells: [0212] 4> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0213] 3> else: [0214] 4> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0215] 4> If the number of cell(s) in the cellsTriggeredList is larger than or equal to numberOfTriggeringCells: [0216] 5> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0217] 5> initiate the measurement reporting procedure; [0218] 2> if the triggerType is set to event and if the leaving condition applicable for this event is fulfilled for one or more of the cells included in the cellsTriggeredList defined within the VarMeasReportList for this measId for all measurements after layer 3 filtering taken during timeToTrigger defined within the VarMeasConfig for this event: [0219] 3> remove the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0220] 3> if reportOnLeave is set to TRUE for the corresponding reporting configuration or if a6-ReportOnLeave is set to TRUE or if a4-a5-ReportOnLeave is set to TRUE for the corresponding reporting configuration: [0221] 4> initiate the measurement reporting procedure; [0222] 3> if the cellsTriggeredList defined within the VarMeasReportList for this measId is empty: [0223] 4> remove the measurement reporting entry within the VarMeasReportList for this measId; [0224] 4> stop the periodical reporting timer for this measId, if running;

[0225] Hereinafter, technical features related to measurement object addition/modification are described. Parts of section 5.5.2.5 and section 5.5.4.1 of 3GPP TS 38.331 v17.0.0 may be referred.

[0226] The UE shall: [0227] 1> for each measObjectId included in the received measObjectToAddModList: [0228] 2> if an entry with the matching measObjectId exists in the measObjectList within the VarMeasConfig, for this entry: [0229] 3> reconfigure the entry with the value received for this measObject, except for the fields cellsToAddModList, excludedCellsToAddModList, allowedCellsToAddModList, cellsToRemoveList, excludedCellsToRemoveList, allowedCellsToRemoveList, tx-PoolMeasToRemoveList, tx-PoolMeasToAddModList, ssb-PositionQCL-CellsToRemoveList, and ssb-PositionQCL-CellsToAddModList; [0230] 3> if the received measObject includes the cellsToRemoveList: [0231] 4> for each physCellId included in the cellsToRemoveList: [0232] 5> remove the entry with the matching physCellId from the cellsToAddModList; [0233] 3> if the received measObject includes the cellsToAddModList: [0234] 4> for each physCellId value included in the cellsToAddModList: [0235] 5> if an entry with the matching physCellId exists in the cellsToAddModList: [0236] 6> replace the entry with the value received for this physCellId; [0237] 5> else: [0238] 6> add a new entry for the received physCellId to the cellsToAddModList; [0239] 3> if the received measObject includes the excludedCellsToRemoveList: [0240] 4> for each pci-RangeIndex included in the excludedCellsToRemoveList: [0241] 5> remove the entry with the matching pci-RangeIndex from the excludedCellsToAddModList; [0242] For each pci-RangeIndex included in the excludedCellsToRemoveList that concerns overlapping ranges of cells, a cell is removed from the exclude-list of cells only if all PCI ranges containing it are removed. [0243] 3> if the received measObject includes the excludedCellsToAddModList: [0244] 4> for each pci-RangeIndex included in the excludedCellsToAddModList: [0245] 5> if an entry with the matching pci-RangeIndex is included in the excludedCellsToAddModList: [0246] 6> replace the entry with the value received for this pci-RangeIndex; [0247] 5> else: [0248] 6> add a new entry for the received pci-RangeIndex to the excludedCellsToAddModList; [0249] 3> if the received measObject includes the allowedCellsToRemoveList: [0250] 4> for each pci-RangeIndex included in the allowedCellsToRemoveList: [0251] 5> remove the entry with the matching pci-RangeIndex

from the allowedCellsToAddModList; [0252] For each pci-RangeIndex included in the allowedCellsToRemoveList that concerns overlapping ranges of cells, a cell is removed from the allow-list of cells only if all PCI ranges containing it are removed. [0253] 3> if the received measObject includes the allowedCellsToAddModList: [0254] 4> for each pci-RangeIndex included in the allowedCellsToAddModList: [0255] 5> if an entry with the matching pci-RangeIndex is included in the allowedCellsToAddModList: [0256] 6> replace the entry with the value received for this pci-RangeIndex; [0257] 5> else: [0258] 6> add a new entry for the received pci-RangeIndex to the allowedCellsToAddModList [0259] 3> for each measId associated with this measObjectId in the measIdList within the VarMeasConfig, if any: [0260] 4> remove the measurement reporting entry for this measId from the VarMeasReportList, if included; [0261] 4> stop the periodical reporting timer or timer T321 or timer T322, whichever one is running, and reset the associated information (e.g. timeToTrigger) for this measId; [0262] 3> if the received measObject includes the tx-PoolMeasToRemoveList: [0263] 4> for each transmission resource pool indicated in tx-PoolMeasToRemoveList: [0264] 5> remove the entry with the matching identity of the transmission resource pool from the tx-PoolMeasToAddModList, [0265] 3> if the received measObject includes the tx-PoolMeasToAddModList: [0266] 4> for each transmission resource pool indicated in tx-PoolMeasToAddModList: [0267] 5> if an entry with the matching identity of the transmission resource pool exists in the tx-PoolMeasToAddModList: [0268] 6> replace the entry with the value received for this transmission resource pool; [0269] 5> else: [0270] 6> add a new entry for the received identity of the transmission resource pool to the tx-PoolMeasToAddModList; [0271] 3> if the received measObject includes the ssb-PositionQCL-CellsToRemoveList: [0272] 4> for each physCellId included in the ssb-PositionQCL-CellsToRemoveList: [0273] 5> remove the entry with the matching physCellId from the ssb-PositionQCL-CellsToAddModList, [0274] 3> if the received measObject includes the ssb-PositionQCL-CellsToAddModList: [0275] 4> for each physCellId included in the ssb-PositionQCL-CellsToAddModList: [0276] 5> if an entry with the matching physCellId exists in the ssb-PositionQCL-CellsToAddModList: [0277] 6> replace the entry with the value received for this physCellId; [0278] 5> else: [0279] 6> add a new entry for the received physCellId to the ssb-PositionQCL-CellsToAddModList; [0280] 2> else: [0281] 3> add a new entry for the received measObject to the measObjectList within VarMeasConfig.

[0282] If AS security has been activated successfully, the UE shall: [0283] 1> for each measId included in the measIdList within VarMeasConfig: [0284] 2> if the corresponding reportConfig includes a reportType set to eventTriggered or periodical: [0285] 3> if the corresponding measObject concerns NR: [0286] 4> if the corresponding reportConfig includes measRSSI-ReportConfig: [0287] 5> consider the resource indicated by the rmtc-Config on the associated frequency to be applicable; [0288] 4> if the eventA1 or eventA2 is configured in the corresponding reportConfig: [0289] 5> consider only the serving cell to be applicable; [0290] 4> if the eventA3 or eventA5 is configured in the corresponding reportConfig: [0291] 5> if a serving cell is associated with a measObjectNR and neighbours are associated with another measObjectNR, consider any serving cell associated with the other measObjectNR to be a neighbouring cell as well; [0292] 4> if corresponding reportConfig includes reportType set to periodical; or [0293] 4> for measurement events other than eventA1 or eventA2: [0294] 5> if useAllowedCellList is set to true: [0295] 6> consider any neighbouring cell detected based on parameters in the associated measObjectNR to be applicable when the concerned cell is included in the allowedCellsToAddModList defined within the VarMeasConfig for this measId; [0296] 5> else: [0297] 6> consider any neighbouring cell detected based on parameters in the associated measObjectNR to be applicable when the concerned cell is not included in the excludedCellsToAddModList defined within the VarMeasConfig for this measId; [0298] 3> else if the corresponding measObject concerns E-UTRA: [0299] 4> if eventB1 or eventB2 is configured in the corresponding reportConfig: [0300] 5> consider a serving cell, if any, on the associated E-UTRA frequency as neighbour cell; [0301] 4> consider any neighbouring

cell detected on the associated frequency to be applicable when the concerned cell is not included in the excludedCellsToAddModListEUTRAN defined within the VarMeasConfig for this measId; [0302] 3> else if the corresponding measObject concerns UTRA-FDD: [0303] 4> if eventB1-UTRA-FDD or eventB2-UTRA-FDD is configured in the corresponding reportConfig; or [0304] 4> if corresponding reportConfig includes reportType set to periodical: [0305] 5> consider a neighbouring cell on the associated frequency to be applicable when the concerned cell is included in the cellsToAddModList defined within the VarMeasConfig for this measId; [0306] 3> else if the corresponding measObject concerns L2 U2N Relay UE: [0307] 4> if eventY1-Relay is configured in the corresponding reportConfig; or [0308] 4> if corresponding reportConfig includes reportType set to periodical: [0309] 5> consider any L2 U2N Relay UE detected on the associated frequency to be applicable for this measId; [0310] 2> else if the corresponding reportConfig includes a reportType set to reportCGI: [0311] 3> consider the cell detected on the associated measObject which has a physical cell identity matching the value of the cellForWhichToReportCGI included in the corresponding reportConfig within the VarMeasConfig to be applicable; [0312] 2> else if the corresponding reportConfig includes a reportType set to reportSFTD: [0313] 3> if the corresponding measObject concerns NR: [0314] 4> if the reportSFTD-Meas is set to true: [0315] 5> consider the NR PSCell to be applicable; [0316] 4> else if the reportSFTD-NeighMeas is included: [0317] 5> if cellsForWhichToReportSFTD is configured in the corresponding reportConfig: [0318] 6> consider any NR neighbouring cell detected on the associated measObjectNR which has a physical cell identity that is included in the cellsForWhichToReportSFTD to be applicable; [0319] 5> else: [0320] 6> consider up to 3 strongest NR neighbouring cells detected based on parameters in the associated measObjectNR to be applicable when the concerned cells are not included in the excludedCellsToAddModList defined within the VarMeasConfig for this measId; [0321] 3> else if the corresponding measObject concerns E-UTRA: [0322] 4> if the reportSFTD-Meas is set to true: [0323] 5> consider the E-UTRA PSCell to be applicable; [0324] 2> else if the corresponding reportConfig includes a reportType set to cli-Periodical or cli-EventTriggered: [0325] 3> consider all CLI measurement resources included in the corresponding measObject to be applicable; [0326] 2> else if the corresponding reportConfig includes a reportType set to rxTxPeriodical: [0327] 3> consider all Rx-Tx time difference measurement resources included in the corresponding measObject to be applicable; [0328] 2> if the corresponding reportConfig concerns the reporting for NR sidelink communication (i.e. reportConfigNR-SL): [0329] 3> consider the transmission resource pools indicated by the tx-PoolMeasToAddModList defined within the VarMeasConfig for this measId to be applicable; [0330] 2> if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable cells for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig, while the VarMeasReportList does not include a measurement reporting entry for this measId (a first cell triggers the event): [0331] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0332] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0333] 3> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0334] 3> if useT312 is set to true in reportConfig for this event: [0335] 4> if T310 for the corresponding SpCell is running; and [0336] 4> if T312 is not running for corresponding SpCell: [0337] 5> start timer T312 for the corresponding SpCell with the value of T312 configured in the corresponding measObjectNR; [0338] 3> initiate the measurement reporting procedure; [0339] 2> else if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable cells not included in the cellsTriggeredList for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the

VarMeasConfig (a subsequent cell triggers the event): [0340] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0341] 3> include the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0342] 3> if useT312 is set to true in reportConfig for this event: [0343] 4> if T310 for the corresponding SpCell is running; and [0344] 4> if T312 is not running for corresponding SpCell: [0345] 5> start timer T312 for the corresponding SpCell with the value of T312 configured in the corresponding measObjectNR; [0346] 3> initiate the measurement reporting procedure; [0347] 2> else if the reportType is set to eventTriggered and if the leaving condition applicable for this event is fulfilled for one or more of the cells included in the cellsTriggeredList defined within the VarMeasReportList for this measId for all measurements after layer 3 filtering taken during timeToTrigger defined within the VarMeasConfig for this event: [0348] 3> remove the concerned cell(s) in the cellsTriggeredList defined within the VarMeasReportList for this measId; [0349] 3> if reportOnLeave is set to true for the corresponding reporting configuration: [0350] 4> initiate the measurement reporting procedure; [0351] 3> if the cellsTriggeredList defined within the VarMeasReportList for this measId is empty: [0352] 4> remove the measurement reporting entry within the VarMeasReportList for this measId; [0353] 4> stop the periodical reporting timer for this measId, if running; [0354] 2> if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable L2 U2N Relay UEs for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig, while the VarMeasReportList does not include a measurement reporting entry for this measId (a first L2 U2N Relay UE triggers the event): [0355] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0356] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0357] 3> include the concerned L2 U2N Relay UE(s) in the relaysTriggeredList defined within the VarMeasReportList for this measId; [0358] 3> initiate the measurement reporting procedure; [0359] 2> else if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable L2 U2N Relay UEs not included in the relaysTriggeredList for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig (a subsequent L2 U2N Relay UE triggers the event): [0360] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0361] 3> include the concerned L2 U2N Relay UE(s) in the relays TriggeredList defined within the VarMeasReportList for this measId; [0362] 3> initiate the measurement reporting procedure; [0363] 2> else if the reportType is set to eventTriggered and if the leaving condition applicable for this event is fulfilled for one or more of the L2 U2N Relay UEs included in the relaysTriggeredList defined within the VarMeasReportList for this measId for all measurements after layer 3 filtering taken during timeToTrigger defined within the VarMeasConfig for this event: [0364] 3> remove the concerned L2 U2N Relay UE(s) in the relaysTriggeredList defined within the VarMeasReportList for this measId; [0365] 3> if reportOnLeave is set to true for the corresponding reporting configuration: [0366] 4> initiate the measurement reporting procedure; [0367] 3> if the relaysTriggeredList defined within the VarMeasReportList for this measId is empty: [0368] 4> remove the measurement reporting entry within the VarMeasReportList for this measId; [0369] 4> stop the periodical reporting timer for this measId, if running; [0370] 2> else if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable transmission resource pools for all measurements taken during timeToTrigger defined for this event within the VarMeasConfig, while the VarMeasReportList does not include a measurement reporting entry for this measId (a first transmission resource pool triggers the event): [0371] 3> include a measurement reporting entry within the VarMeasReportList for this measId;

[0372] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0373] 3> include the concerned transmission resource pool(s) in the poolsTriggeredList defined within the VarMeasReportList for this measId; [0374] 3> initiate the measurement reporting procedure; [0375] 2> else if the reportType is set to eventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable transmission resource pools not included in the poolsTriggeredList for all measurements taken during timeToTrigger defined for this event within the VarMeasConfig (a subsequent transmission resource pool triggers the event): [0376] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0377] 3> include the concerned transmission resource pool(s) in the poolsTriggeredList defined within the VarMeasReportList for this measId; [0378] 3> initiate the measurement reporting procedure; [0379] 2> else if the reportType is set to eventTriggered and if the leaving condition applicable for this event is fulfilled for one or more applicable transmission resource pools included in the poolsTriggeredList defined within the VarMeasReportList for this measId for all measurements taken during timeToTrigger defined within the VarMeasConfig for this event: [0380] 3> remove the concerned transmission resource pool(s) in the poolsTriggeredList defined within the VarMeasReportList for this measId; [0381] 3> if the poolsTriggeredList defined within the VarMeasReportList for this measId is empty: [0382] 4> remove the measurement reporting entry within the VarMeasReportList for [0383] this measId; [0384] 4> stop the periodical reporting timer for this measId, if running [0385] 2> else if the reportType is set to eventTriggered and if the eventId is set to eventD1 and if the entering condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled during timeToTrigger defined within the VarMeasConfig for this event: [0386] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0387] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0388] 3> initiate the measurement reporting procedure; [0389] 2> if reportType is set to periodical and if a (first) measurement result is available: [0390] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0391] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0392] 3> if the corresponding reportConfig includes measRSSI-ReportConfig: [0393] 4> initiate the measurement reporting procedure immediately when RSSI sample values are reported by the physical layer after the first L1 measurement duration; [0394] 3> else if the corresponding reportConfig includes the ul-DelayValueConfig: [0395] 4> initiate the measurement reporting procedure immediately after a first measurement result is provided from lower layers of the associated DRB identity; [0396] 3> else if the corresponding reportConfig includes the ul-ExcessDelayConfig: [0397] 4> initiate the measurement reporting procedure immediately after a first measurement result is provided from lower layers of the associated DRB identity (ies) according to the configured threshold per DRB identity (ies); [0398] 3> else if the reportAmount exceeds 1: [0399] 4> initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for the NR SpCell or for the serving L2 U2N Relay UE (if the UE is a L2 U2N Remote UE); [0400] 3> else (i.e. the reportAmount is equal to 1): [0401] 4> initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for the NR SpCell and for the strongest cell among the applicable cells, or for the NR SpCell and for the strongest L2 U2N Relay UEs among the applicable L2 U2N Relay UEs; or initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for the serving L2 U2N Relay UE and for the strongest cell among the applicable cells (if the UE is a L2 U2N Remote UE); [0402] 2> if, in case the corresponding reportConfig concerns the reporting for NR sidelink communication, reportType is set to periodical and if a (first) measurement result is available: [0403] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0404] 3> set the numberOfReportsSent defined within the

VarMeasReportList for this measId to 0; [0405] 3> initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for the NR SpCell and CBR measurement results become available; [0406] 2> if the reportType is set to cli-EventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more applicable CLI measurement resources for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig, while the VarMeasReportList does not include a measurement reporting entry for this measId (a first CLI measurement resource triggers the event): [0407] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0408] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0409] 3> include the concerned CLI measurement resource(s) in the cli-TriggeredList defined within the VarMeasReportList for this measId; [0410] 3> initiate the measurement reporting procedure; [0411] 2> else if the reportType is set to cli-EventTriggered and if the entry condition applicable for this event, i.e. the event corresponding with the eventId of the corresponding reportConfig within VarMeasConfig, is fulfilled for one or more CLI measurement resources not included in the cli-TriggeredList for all measurements after layer 3 filtering taken during timeToTrigger defined for this event within the VarMeasConfig (a subsequent CLI measurement resource triggers the event): [0412] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0413] 3> include the concerned CLI measurement resource(s) in the cli-TriggeredList defined within the VarMeasReportList for this measId; [0414] 3> initiate the measurement reporting procedure; [0415] 2> else if the reportType is set to cli-EventTriggered and if the leaving condition applicable for this event is fulfilled for one or more of the CLI measurement resources included in the cli-TriggeredList defined within the VarMeasReportList for this measId for all measurements after layer 3 filtering taken during timeToTrigger defined within the VarMeasConfig for this event: [0416] 3> remove the concerned CLI measurement resource(s) in the cli-TriggeredList defined within the VarMeasReportList for this measId; [0417] 3> if reportOnLeave is set to true for the corresponding reporting configuration: [0418] 4> initiate the measurement reporting procedure; [0419] 3> if the cli-TriggeredList defined within the VarMeasReportList for this measId is empty: [0420] 4> remove the measurement reporting entry within the VarMeasReportList for this measId; [0421] 4> stop the periodical reporting timer for this measId, if running; [0422] 2> if reportType is set to cli-Periodical and if a (first) measurement result is available: [0423] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0424] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0425] 3> initiate the measurement reporting procedure, immediately after the quantity to be reported becomes available for at least one CLI measurement resource; [0426] 2> if reportType is set to rxTxPeriodical and if a (first) measurement result is available: [0427] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0428] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0429] 3> initiate the measurement reporting procedure; [0430] 2> upon expiry of the periodical reporting timer for this measId: [0431] 3> initiate the measurement reporting procedure. [0432] 2> if the corresponding reportConfig includes a reportType is set to reportSFTD: [0433] 3> if the corresponding measObject concerns NR: [0434] 4> if the drx-SFTD-NeighMeas is included: [0435] 5> if the quantity to be reported becomes available for each requested pair of PCell and NR cell: [0436] 6> stop timer T322; [0437] 6> initiate the measurement reporting procedure; [0438] 4> else [0439] 5> initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for each requested pair of PCell and NR cell or the maximal measurement reporting delay; [0440] 3> else if the corresponding measObject concerns E-UTRA: [0441] 4> initiate the measurement reporting procedure immediately after the quantity to be reported becomes available for the pair of PCell and E-UTRA PSCell or the maximal measurement reporting delay; [0442] 2> if reportType is set to reportCGI: [0443] 3> if the UE

acquired the SIB1 or SystemInformationBlockType1 for the requested cell; or [0444] 3> if the UE detects that the requested NR cell is not transmitting SIB1: [0445] 4> stop timer T321; [0446] 4> include a measurement reporting entry within the VarMeasReportList for this measId; [0447] 4> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0448] 4> initiate the measurement reporting procedure; [0449] 2> upon the expiry of T321 for this measId: [0450] 3> include a measurement reporting entry within the VarMeasReportList for this measId; [0451] 3> set the numberOfReportsSent defined within the VarMeasReportList for this measId to 0; [0452] 3> initiate the measurement reporting procedure. [0453] 2> upon the expiry of T322 for this measId: [0454] 3> initiate the measurement reporting procedure.

[0455] Meanwhile, as the altitude of aerial UEs increases, they experience a line-of-sight propagation condition to more cells, making faraway cells more visible. Consequently, aerial UEs receive interference from more cells in the downlink and cause interference to more cells in the uplink. Additionally, aerial coverage becomes fragmented at higher altitudes. In contrast to terrestrial UEs, which are typically served by the closest network, aerial UEs are served by a side lobe of a neighboring network far from the UE. Therefore, the faraway network may become the serving network, resulting in non-continuous cell coverage.

[0456] As an aerial UE flies across multiple cells at a specific height and direction, the quality of its serving cell fluctuates with a certain frequency, depending on network evaluation. Furthermore, aerial UEs can observe multiple cells with similar signal strength, making it challenging to select a suitable cell based on cell quality alone, such as RSRP and RSRQ. In the case of the measurement report, if an aerial UE detects multiple neighbor cells, it may send measurement reports more frequently, causing more signaling overhead and air interference. To mitigate this issue, the network can configure an excluded list or allowed list of cells to restrict the report. If a cell is on the excluded cell list, the UE cannot send a measurement report for that cell. If the allowed cell list is configured, the UE can send a measurement report for only cells belonging to the allowed cell list.

[0457] However, to configure the excluded cell list or allowed cell list based on height, the UE should send a measurement report based on the height (H1/H2 event), allowing the network to send a new cell configuration based on the UE's current height. This also results in more signaling overhead between the UE and the network.

[0458] Therefore, studies for measurement report considering height in a wireless communication system are required.

[0459] Hereinafter, a method for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described with reference to the following drawings.

[0460] The following drawings are created to explain specific embodiments of the present disclosure. The names of the specific devices or the names of the specific signals/messages/fields shown in the drawings are provided by way of example, and thus the technical features of the present disclosure are not limited to the specific names used in the following drawings. Herein, a wireless device may be referred to as a user equipment (UE).

[0461] FIG. **10** shows an example of a method for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure.

[0462] In particular, FIG. **10** shows an example of a method performed by a wireless device in a wireless communication system.

[0463] In step **S1001**, a wireless device may receive information on a cell list associated with a height range.

[0464] For example, the cell list may include an allowed cell list and/or an excluded cell list. For example, the cell list may be an allowed cell list and/or an excluded cell list.

[0465] In other words, the wireless device may receive information on an allowed cell list and/or an excluded cell list.

[0466] For example, the height range may be indicated by a maximum height and/or a minimum height. For example, the information on the cell list associated with the height range may include information on one or more cell lists and one or more the height ranges associated with each cell list. The associated height ranges may be indicated by the maximum height and/or the minimum height (or threshold heights).

[0467] For example, the cell list may be configured per frequency.

[0468] For example, the information on the cell list associated with the height range may be included in a measurement configuration. For example, the cell list may be included in the measurement objects of the measurement configuration. The information on the associated height range may be included in the measurement configuration.

[0469] For example, the measurement configuration may be associated with the height range.

[0470] In step **S1002**, a wireless device may monitor a current height of the wireless device.

[0471] For example, the wireless device may determine a current height range among the plurality of height ranges based on the height of the wireless device.

[0472] The wireless device may apply a configuration associated with the current height range. For example, the configuration associated with the current height range may include a measurement configuration.

[0473] For example, when the current height range is different from the previous height range, the wireless device may apply a measurement configuration associated with the current height range. In other words, then the height range of the wireless device is changed, the wireless device may apply a configuration associated with the current height range.

[0474] In step **S1003**, a wireless device may set a measurement report while the current height of the wireless device is in the height range.

[0475] For example, the wireless device may perform measurements on the frequencies in the measurement objects included in the measurement configuration.

[0476] For example, the wireless device may perform measurements based on a measurement configuration.

[0477] For example, the wireless device may only perform measurements on the frequencies and/or cells included in the allowed cell list. The wireless device may not the frequencies and/or cells included in the excluded cell list.

[0478] For another example, the wireless device may perform measurements on the frequencies and/or cells included in the allowed cell list and/or the excluded cell list. The wireless device may determine measurement report to include only the measurement results on a certain frequency and/or a certain cell that is included in the allowed cell list or not included in the excluded cell list, in step **S1004** below.

[0479] In step **S1004**, a wireless device may determine whether the measurement report includes measurement results of a first cell based on the cell list.

[0480] For example, a wireless device may determine that the measurement report includes measurement results of the first cell based on the first cell being included in an allowed cell list.

[0481] For example, a wireless device may determine that the measurement report includes measurement results of the first cell based on the first cell not being included in an excluded cell list.

[0482] For example, a wireless device may determine that the measurement report does not include measurement results of a second cell based on the second cell being included in an excluded cell list.

[0483] For example, a wireless device may determine that the measurement report does not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

[0484] In step **S1005**, a wireless device may transmit the measurement report.

[0485] For example, the wireless device may transmit the measurement report to a network, based

on at least one condition for the transmission of the measurement report is satisfied.

[0486] According to some embodiments of the present disclosure, the wireless device may receive information on a first cell list associated with a first height range and a second cell list associated with a second height range. The first cell may be included in the first cell list. The second cell may be included in the second cell list.

[0487] Based on the current height of the wireless device being in the first height range: the wireless device may perform measurements on the first cell and the second cell. The wireless device may set the measurement report to include measurement results of the first cell but not to include measurement results of the second cell. In this case, the first cell list and the second cell list may be allowed cell lists, respectively.

[0488] Based on the current height of the wireless device being in the second height range: the wireless device may perform measurements on the first cell and the second cell. The wireless device may set the measurement report to include measurement results of the second cell but not to include measurement results of the first cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0489] According to some embodiments of the present disclosure, the wireless device may be in communication with at least one of a user equipment, a network, or an autonomous vehicle other than the wireless device.

[0490] For example, the wireless device is a mobile device capable of vertical mobility.

[0491] Hereinafter, some embodiments of a method for measurement report considering height in a wireless communication system are described.

[0492] The network may configure a list of excluded cells for which the UE cannot send a measurement report message for the cells included in the list.

[0493] The network may configure a list of allowed cells for which the UE can send a measurement report message only for the cells included in the list.

[0494] For example, the network may configure applicable height information for the allowed or excluded cell list. [0495] For example, the network may configure one or more height ranges for

which the excluded cell list is applied. The network may configure one or more height ranges for which the allowed cell list is applied. [0496] The height information may be configured for each

allowed cell list and/or for each excluded cell list. The height information may be configured commonly for multiple lists. [0497] Height information may be expressed by one or more height

thresholds. [0498] The height information may be used to divide an entire height space into multiple height range sections to apply or not to apply the associated excluded/allowed cell list.

[0499] The height information may indicate allowed height range. The height information may

indicate non-allowed height range. [0500] The height information may indicate a height range to distinguish available cells in the same excluded/allowed cell list. There are three options for

configuring an allowed height. [0501] 1) The height information may indicate a maximum height for a cell to control measurement report triggered by the cell. [0502] 2) The height information may

indicate a minimum height for a cell to control measurement report triggered by the cell. [0503] 3) The height information may indicate a height boundary expressed by min height and max height to

control measurement report triggered by the cell.

[0504] For example, UE determines its current height.

[0505] UE determines applicable allowed cell list and excluded cell list based on the current height and the configured height range information. [0506] For example, network may configure allowed

and excluded cell list for each frequency and height range information as follows: [0507] >

Frequency1 [0508] >> Allowed cell list is configured [0509] >> Non-allowed cell list is configured

[0510] >> Non-allowed height range3 for allowed cell list [0511] >> Non-allowed height range3

for excluded cell list [0512] > Frequency2 [0513] >> Allowed cell list is configured [0514] >>

Non-allowed cell list is configured [0515] >> Allowed height range2 for allowed cell list [0516] >>

Non-allowed height range1 for excluded cell list [0517] > Frequency3 [0518] >> Allowed cell list

is configured [0519] >> Non-allowed cell list is configured [0520] >> No height range for allowed cell list [0521] >> No height range for excluded cell list

[0522] The, the UE determines whether each allowed cell list and excluded cell list is applicable.

[0523] Table 5 shows an example of the applicable allowed cell list and excluded cell list.

TABLE-US-00005 TABLE 5 Height range1 Height range2 Height range3 Frequency1 allowed
Applicable Applicable Non-applicable cell list Frequency1 Applicable Applicable Non-applicable
excluded cell list Frequency2 allowed Non-applicable Applicable Non-applicable cell list
Frequency2 Non-applicable Applicable Applicable excluded cell list Frequency3 allowed
Applicable Applicable Applicable cell list Frequency3 Applicable Applicable Applicable excluded
cell list

[0524] The UE may send measurement report based on the applicable excluded cell list and applicable allowed cell list.-Any cell in the applicable excluded cell list, if configured, cannot trigger measurement report triggering, i.e., such cell is not included in cellTriggeredList. [0525] A cell in the non-applicable excluded cell list, if configured, may trigger measurement report triggering if the cell satisfies measurement reporting condition. [0526] Any cell in the non-applicable allowed cell list, if configured, cannot trigger measurement report, i.e., such cell is not included in cellTriggeredList. [0527] A cell in the applicable excluded cell list, if configured, may trigger measurement report triggering if the cell satisfies measurement reporting condition.

[0528] According to some embodiments of the present disclosure, a wireless device may receive a measurement configuration of a cell. The configuration may include an excluded cell list or an allowed cell list. Each list is associated with a range of height.

[0529] A wireless device may perform measurement for the cell. A wireless device may determine a current height.

[0530] A wireless device may send measurement report based on the determined height if measurement result of the cell satisfies the report condition. For example, the cell is not included in excluded cell list, or the cell is included in allowed cell list.

[0531] FIG. 11 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0532] In particular, FIG. 11 illustrates a case of an excluded Cell list with a height threshold. In FIG. 11, an allowed height may be configured to divide a section to apply a different excluded cell list. [0533] (1) The network configures a first excluded cell list and a second excluded cell list with a height threshold. [0534] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0535] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the first excluded cell list. [0536] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the second cell list.

[0537] FIG. 12 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0538] In particular, FIG. 12 illustrates a case of an excluded Cell list with a minimum height threshold. In FIG. 12, an allowed height is configured to each cell in an excluded cell list. [0539] (1) The network configures an excluded cell list, and the cells in the excluded cell list may be associated with a minimum height threshold. [0540] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0541] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list and is available in the excluded cell list at the current height.

[0542] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list.

[0543] FIG. 13 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0544] In particular, FIG. 13 illustrates a case of an excluded Cell list with a maximum height threshold. In FIG. 13, an allowed height is configured to each cell in an excluded cell list. [0545] (1) The network configures an excluded cell list, and the cells in the excluded cell list may be associated with a maximum height threshold. [0546] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0547] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list. [0548] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list and is available in the excluded cell list at the current height.

[0549] FIG. 14 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0550] In particular, FIG. 14 illustrates a case of an excluded Cell list with a height boundary. In FIG. 14, a height boundary is configured for the section to apply a different excluded cell list.

[0551] (1) The network configures a first excluded cell list and a second excluded cell list with a height boundary. The height is configured with a minimum threshold and a maximum threshold.

[0552] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0553] (3) At an altitude lower than the minimum threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the first excluded cell list. [0554] (4) At an altitude between the minimum threshold and the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the second excluded cell list. [0555] (5) At an altitude higher than the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the first excluded cell list.

[0556] FIG. 15 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0557] In particular, FIG. 15 illustrates a case of an excluded Cell list with a height boundary. In FIG. 15, a height boundary is configured to each cell in an excluded cell list. [0558] (1) The

network configures an excluded cell list, and the cells in the excluded cell list may be associated with a height boundary. The height is configured with a minimum threshold and a maximum threshold. [0559] (2) The UE performs measurement based on a measurement configuration, which

includes measurement object(s) and report configurations, for a cell. [0560] (3) At an altitude lower than the minimum threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list. [0561] (4) At an altitude between the minimum threshold and the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE cannot send a measurement report for the cell if the cell is included in the excluded cell list and is available in the excluded cell at the current height. [0562] (5) At an altitude higher than the maximum threshold, when the measurement result of the cell satisfies the report configuration, the

UE cannot send a measurement report for the cell if the cell is included in the excluded cell list. [0563] FIG. 16 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0564] In particular, FIG. 16 illustrates a case of an allowed Cell list with a height threshold. In FIG. 16, an allowed height is configured to divide a section to apply a different allowed cell list.

[0565] (1) The network configures a first allowed cell list and a second allowed cell list with a height threshold. [0566] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0567] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the first allowed cell list. [0568] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the second allowed cell list.

[0569] FIG. 17 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0570] In particular, FIG. 17 illustrates a case of an allowed Cell list with a minimum height threshold. In FIG. 17, an allowed height is configured to each cell in an allowed cell list. [0571] (1) The network configures an allowed cell list, and the cells in the allowed cell list may be associated with a minimum height threshold. [0572] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0573] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list and is available in the allowed cell list at the current height. [0574] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list.

[0575] FIG. 18 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0576] In particular, FIG. 18 illustrates a case of an Allowed Cell list with a maximum height threshold. In FIG. 18, an allowed height is configured to each cell in an allowed cell list. [0577] (1) The network configures an allowed cell list, and the cells in the allowed cell list may be associated with a maximum height threshold. [0578] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0579] (3) At an altitude lower than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list. [0580] (4) At an altitude higher than the threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list and is available in the allowed cell list at the current height.

[0581] FIG. 19 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0582] In particular, FIG. 19 illustrates a case of an allowed Cell list with a height boundary. In FIG. 19, a height boundary is configured for the section to apply a different allowed cell list. [0583]

(1) The network configures a first allowed cell list and a second allowed cell list with a height boundary. The height is configured with a minimum threshold and a maximum threshold. [0584]

(2) The UE performs measurement based on a measurement configuration, which includes

measurement object(s) and report configurations, for a cell. [0585] (3) At an altitude lower than the minimum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the first allowed cell list. [0586] (4) At an altitude between the minimum threshold and the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the second allowed cell list. [0587] (5) At an altitude higher than the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the first allowed cell list.

[0588] FIG. 20 shows an example of a method for measurement report considering height procedure in a wireless communication system, according to some embodiments of the present disclosure.

[0589] In particular, FIG. 20 illustrates a case of an allowed Cell list with a height boundary. In FIG. 20, a height boundary is configured to each cell in an allowed cell list. [0590] (1) The network configures an allowed cell list, and the cells in the allowed cell list may be associated with a height boundary. The height is configured with a minimum threshold and a maximum threshold. [0591] (2) The UE performs measurement based on a measurement configuration, which includes measurement object(s) and report configurations, for a cell. [0592] (3) At an altitude lower than the minimum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list. [0593] (4) At an altitude between the minimum threshold and the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list and is available in the allowed cell list at the current height.

[0594] (5) At an altitude higher than the maximum threshold, when the measurement result of the cell satisfies the report configuration, the UE can send a measurement report for the cell if the cell is included in the allowed cell list.

[0595] Some of the detailed steps shown in the examples of FIGS. 10-20 may not be essential steps and may be omitted. In addition to the steps shown in FIGS. 10-20, other steps may be added, and the order of the steps may vary. Some of the above steps may have their own technical meaning.

[0596] Hereinafter, an apparatus for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described. Herein, the apparatus may be a wireless device (100 or 200) in FIGS. 2, 3, and 5.

[0597] For example, a wireless device may perform the methods described above. The detailed description overlapping with the above-described contents could be simplified or omitted.

[0598] Referring to FIG. 5, a wireless device 100 may include a processor 102, a memory 104, and a transceiver 106.

[0599] According to some embodiments of the present disclosure, the processor 102 may be configured to be coupled operably with the memory 104 and the transceiver 106.

[0600] The processor 102 may be configured to control the transceiver 106 to receive information on a cell list associated with a height range. The processor 102 may be configured to monitor a current height of the wireless device. The processor 102 may be configured to set a measurement report while the current height of the wireless device is in the height range. The processor 102 may be configured to determine whether the measurement report includes measurement results of a first cell based on the cell list. The processor 102 may be configured to control the transceiver 106 to transmit the measurement report.

[0601] For example, the measurement report may include measurement results of the first cell based on the first cell being included in an allowed cell list.

[0602] For example, the measurement report may include measurement results of the first cell based on the first cell not being included in an excluded cell list.

[0603] For example, the measurement report may not include measurement results of a second cell based on the second cell being included in an excluded cell list.

[0604] For example, the measurement report may not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

[0605] For example, the cell list may include an allowed cell list and/or an excluded cell list.

[0606] For example, the processor **102** may be configured to control the transceiver **106** to receive information on a first cell list associated with a first height range and a second cell list associated with a second height range. A first cell may be included in the first cell list and a second cell may be included in the second cell list.

[0607] For example, based on the current height of the wireless device being in the first height range, the processor **102** may be configured to control the transceiver **106** to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the first cell but not to include measurement results of the second cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0608] For example, based on the current height of the wireless device being in the second height range, the processor **102** may be configured to control the transceiver **106** to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the second cell but not to include measurement results of the first cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0609] For example, the wireless device may be a mobile device capable of vertical mobility.

[0610] For example, the height range may be indicated by a maximum height and/or a minimum height.

[0611] For example, the cell list may be configured per frequency.

[0612] For example, the information on the cell list associated with the height range may be included in a measurement configuration.

[0613] For example, the processor **102** may be configured to control the transceiver **106** to be in communication with at least one of a user equipment, a network, or an autonomous vehicle other than the wireless device.

[0614] Hereinafter, a processor for a wireless device for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described.

[0615] The processor may be configured to control the wireless device to receive information on a cell list associated with a height range. The processor may be configured to control the wireless device to monitor a current height of the wireless device. The processor may be configured to control the wireless device to set a measurement report while the current height of the wireless device is in the height range. The processor may be configured to control the wireless device to determine whether the measurement report includes measurement results of a first cell based on the cell list. The processor may be configured to control the wireless device to transmit the measurement report.

[0616] For example, the measurement report may include measurement results of the first cell based on the first cell being included in an allowed cell list.

[0617] For example, the measurement report may include measurement results of the first cell based on the first cell not being included in an excluded cell list.

[0618] For example, the measurement report may not include measurement results of a second cell based on the second cell being included in an excluded cell list.

[0619] For example, the measurement report may not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

[0620] For example, the cell list may include an allowed cell list and/or an excluded cell list.

[0621] For example, the processor may be configured to control the wireless device to receive information on a first cell list associated with a first height range and a second cell list associated

with a second height range. A first cell may be included in the first cell list and a second cell may be included in the second cell list.

[0622] For example, based on the current height of the wireless device being in the first height range, the processor may be configured to control the wireless device to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the first cell but not to include measurement results of the second cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0623] For example, based on the current height of the wireless device being in the second height range, the processor may be configured to control the wireless device to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the second cell but not to include measurement results of the first cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0624] For example, the wireless device may be a mobile device capable of vertical mobility.

[0625] For example, the height range may be indicated by a maximum height and/or a minimum height.

[0626] For example, the cell list may be configured per frequency.

[0627] For example, the information on the cell list associated with the height range may be included in a measurement configuration.

[0628] For example, the processor may be configured to control the wireless device to be in communication with at least one of a user equipment, a network, or an autonomous vehicle other than the wireless device.

[0629] Hereinafter, a non-transitory computer-readable medium has stored thereon a plurality of instructions for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described.

[0630] According to some embodiment of the present disclosure, the technical features of the present disclosure could be embodied directly in hardware, in a software executed by a processor, or in a combination of the two. For example, a method performed by a wireless device in a wireless communication may be implemented in hardware, software, firmware, or any combination thereof. For example, a software may reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other storage medium.

[0631] Some example of storage medium is coupled to the processor such that the processor can read information from the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. For other example, the processor and the storage medium may reside as discrete components.

[0632] The computer-readable medium may include a tangible and non-transitory computer-readable storage medium.

[0633] For example, non-transitory computer-readable media may include random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, or any other medium that can be used to store instructions or data structures. Non-transitory computer-readable media may also include combinations of the above.

[0634] In addition, the method described herein may be realized at least in part by a computer-readable communication medium that carries or communicates code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer.

[0635] According to some embodiment of the present disclosure, a non-transitory computer-readable medium has stored thereon a plurality of instructions. The stored a plurality of instructions may be executed by a processor of a wireless device.

[0636] The stored a plurality of instructions may cause the wireless device to receive information

on a cell list associated with a height range. The stored a plurality of instructions may cause the wireless device to monitor a current height of the wireless device. The stored a plurality of instructions may cause the wireless device to set a measurement report while the current height of the wireless device is in the height range. The stored a plurality of instructions may cause the wireless device to determine whether the measurement report includes measurement results of a first cell based on the cell list. The stored a plurality of instructions may cause the wireless device to transmit the measurement report.

[0637] For example, the measurement report may include measurement results of the first cell based on the first cell being included in an allowed cell list.

[0638] For example, the measurement report may include measurement results of the first cell based on the first cell not being included in an excluded cell list.

[0639] For example, the measurement report may not include measurement results of a second cell based on the second cell being included in an excluded cell list.

[0640] For example, the measurement report may not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

[0641] For example, the cell list may include an allowed cell list and/or an excluded cell list.

[0642] For example, the stored a plurality of instructions may cause the wireless device to receive information on a first cell list associated with a first height range and a second cell list associated with a second height range. A first cell may be included in the first cell list and a second cell may be included in the second cell list.

[0643] For example, based on the current height of the wireless device being in the first height range, the stored a plurality of instructions may cause the wireless device to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the first cell but not to include measurement results of the second cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0644] For example, based on the current height of the wireless device being in the second height range, the stored a plurality of instructions may cause the wireless device to perform measurements on the first cell and the second cell; and set the measurement report to include measurement results of the second cell but not to include measurement results of the first cell. The first cell list and the second cell list may be allowed cell lists, respectively.

[0645] For example, the wireless device may be a mobile device capable of vertical mobility.

[0646] For example, the height range may be indicated by a maximum height and/or a minimum height.

[0647] For example, the cell list may be configured per frequency.

[0648] For example, the information on the cell list associated with the height range may be included in a measurement configuration.

[0649] According to some embodiments of the present disclosure, the stored a plurality of instructions may cause the wireless device to be in communication with at least one of a user equipment, a network, or an autonomous vehicle other than the wireless device.

[0650] Hereinafter, a method performed by a base station (BS) for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described.

[0651] The BS may provide, to a wireless device, information on a cell list associated with a height range. The BS may receive, from the wireless device, measurement report which is set based on the cell list and a height of the wireless device.

[0652] Hereinafter, a base station (BS) for measurement report considering height in a wireless communication system, according to some embodiments of the present disclosure, will be described.

[0653] The BS may include a transceiver, a memory, and a processor operatively coupled to the transceiver and the memory.

[0654] The processor may be configured to control the transceiver to provide, to a wireless device, information on a cell list associated with a height range. The processor may be configured to control the transceiver to receive, from the wireless device, measurement report which is set based on the cell list and a height of the wireless device.

[0655] The present disclosure can have various advantageous effects.

[0656] According to some embodiments of the present disclosure, a wireless device could efficiently transmit the height-based measurement report by using the height-based cell list.

[0657] For example, if the altitude changes rapidly, the UE can directly perform measurements on the target cell. When the altitude changes in a ping-pong flight mode, it is possible to reduce signaling overhead due to measurement configuration. Additionally, by targeting a network that supports UAVs, measurement reports can be transmitted only about suitable cells at the corresponding height. As a result, unnecessary measurement reports can be reduced, leading to a reduction in UL interference.

[0658] In other words, when the altitude changes rapidly, the UE can directly apply the target cell to perform measurement. When the altitude ping-pongs the flight mode, it is possible to reduce the signaling overhead due to the measurement configuration. Furthermore, by targeting a network that supports UAV, measurement report can be transmitted only about a suitable cell in the corresponding height. Therefore, UL interference can be reduced by reducing unnecessary measurement reports.

[0659] For example, by excluding measurement results for unnecessary cells from the measurement report, the wireless device could save resources.

[0660] According to some embodiments of the present disclosure, a wireless network system could provide an efficient solution for the height-based measurement reporting.

[0661] Advantageous effects which can be obtained through specific embodiments of the present disclosure are not limited to the advantageous effects listed above. For example, there may be a variety of technical effects that a person having ordinary skill in the related art can understand and/or derive from the present disclosure. Accordingly, the specific effects of the present disclosure are not limited to those explicitly described herein, but may include various effects that may be understood or derived from the technical features of the present disclosure.

[0662] Claims in the present disclosure can be combined in a various way. For instance, technical features in method claims of the present disclosure can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims can be combined to be implemented or performed in a method. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in an apparatus. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in a method. Other implementations are within the scope of the following claims.

Claims

1. A method, comprising: receiving, by a wireless device, information on a cell list associated with a height range; monitoring, by the wireless device, a current height of the wireless device; setting, by the wireless device, a measurement report while the current height of the wireless device is in the height range; determining, by the wireless device, whether the measurement report includes measurement results of a first cell based on the cell list; and transmitting, by the wireless device, the measurement report.
2. The method of claim 1, wherein the measurement report includes measurement results of the first cell based on the first cell being included in an allowed cell list.
3. The method of claim 1, wherein the measurement report includes measurement results of the first cell based on the first cell not being included in an excluded cell list.
4. The method of claim 1, wherein the measurement report does not include measurement results of

a second cell based on the second cell being included in an excluded cell list.

5. The method of claim 1, wherein the measurement report does not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

6. The method of claim 1, wherein the cell list includes an allowed cell list and/or an excluded cell list.

7. The method of claim 1, wherein the method further comprises, receiving, by the wireless device, information on a first cell list associated with a first height range and a second cell list associated with a second height range, wherein a first cell is included in the first cell list and a second cell is included in the second cell list.

8. The method of claim 7, wherein the method further comprises, based on the current height of the wireless device being in the first height range: performing, by the wireless device, measurements on the first cell and the second cell; and setting, by the wireless device, the measurement report to include measurement results of the first cell but not to include measurement results of the second cell, wherein the first cell list and the second cell list are allowed cell lists, respectively.

9. The method of claim 7, wherein the method further comprises, based on the current height of the wireless device being in the second height range: performing, by the wireless device, measurements on the first cell and the second cell; and setting, by the wireless device, the measurement report to include measurement results of the second cell but not to include measurement results of the first cell, wherein the first cell list and the second cell list are allowed cell lists, respectively.

10. The method of claim 1, wherein the wireless device is a mobile device capable of vertical mobility.

11. The method of claim 1, wherein the height range is indicated by a maximum height and/or a minimum height.

12. The method of claim 1, wherein the cell list is configured per frequency.

13. The method of claim 1, wherein the information on the cell list associated with the height range is included in a measurement configuration.

14. The method of claim 1, wherein the wireless device is in communication with at least one of a user equipment, a network, or an autonomous vehicle other than the wireless device.

15. A wireless device, comprising: a transceiver; a memory; and at least one processor operatively coupled to the transceiver and the memory, and adapted to: control the transceiver to receive information on a cell list associated with a height range; monitor a current height of the wireless device; set a measurement report while the current height of the wireless device is in the height range; determine whether the measurement report includes measurement results of a first cell based on the cell list; and control the transceiver to transmit the measurement report.

16. The wireless device of claim 15, wherein the measurement report includes measurement results of the first cell based on the first cell being included in an allowed cell list.

17. The wireless device of claim 15, wherein the measurement report includes measurement results of the first cell based on the first cell not being included in an excluded cell list.

18. The wireless device of claim 15, wherein the measurement report does not include measurement results of a second cell based on the second cell being included in an excluded cell list.

19. The wireless device of claim 15, wherein the measurement report does not include measurement results of a second cell based on the second cell not being included in an allowed cell list.

20-31. (canceled)

32. A base station, comprising: a transceiver; a memory; and a processor operatively coupled to the transceiver and the memory, and adapted to: provide, to a wireless device, information on a cell list associated with a height range; and receive, from the wireless device, measurement report which is set based on the cell list and a height of the wireless device.

